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A Comprehensive Approach to Material Handling Equipment Selection and Efficiency Improvement

Insights from a Home Appliance Manufacturing Company Warehouse Case Study

DIMITRIOS OIKONOMOU

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Dimitrios Oikonomou

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En omfattande metod för val av materialhanteringsutrustning och effektivitetsförbättring: Insikter från en fallstudie av ett lager hos ett företag inom hushållsapparatstillverkning

Dimitrios Oikonomou

Godkänt	Examinator	Handledare
2024-10-07	Sofia Ritzén	Susanne Nilsson
	Uppdragsgivare	Kontaktperson
	Konfidentiell	Konfidentiell

Sammanfattning

Att analysera användningen av materialhanteringsutrustning (MHE) i ett lager var huvudsyftet med detta examensarbete, som syftade till att öka den operativa effektiviteten och minska kostnaderna. För att identifiera ineffektivitet och flaskhalsar undersöktes hur den nuvarande MHE användes. Arbetet undersökte även lämpliga tillvägagångssätt för att bestämma storlek och typ av framtida MHE. Denna avhandling genomfördes som en fallstudie där MHE i ett lager i Stockholm, Sverige, som tillhör den nordeuropeiska filialen av Home Appliance Manufacturing Company (HAMC), optimerades.

Forskningen baserades på en analys av de typer, kvantiteter och användningsområden för MHE som HAMC använde, tillsammans med de problem som de anställda upplevde. En litteraturstudie genomfördes för att få en djupare förståelse för de olika metoder som används för att välja MHE och hur dessa problem löses för att förbättra verksamhetens effektivitet. Under denna litteraturstudie identifierades fyra olika metoder för val av MHE och utformning av materialhanteringssystem: konventionell, analytisk, kunskapsbaserad och hybrid. Litteraturgenomgången bidrog också till att fastställa vilka nyckeltal som bör följas upp och hur man kan kompensera för deras frånvaro.

Modifieringar gjordes av den framtida MHE:s sammansättning baserat på effektivitetskriterier för att förbättra prestanda och övergripande effektivitet, minska drift-, underhålls- och leasingkostnader samt effektivisera verksamheten. De viktigaste ändringarna omfattade eliminering av utrustning, systemuppgraderingar, säkerhetsprotokoll, parametermätning, analys av utnyttjandet och ett verktyg för allokering av MHE. Medarbetarnas deltagande i forskningen och utformningen av det nya MHE-systemet tyder på att det kommer att bli mer effektivt, och detta var de viktigaste förändringarna av systemet. Efter att ha bekräftat forskningen och metodiken var projektsponsorerna nöjda med resultaten av det nya materialhanteringssystemets sammansättning. Dessutom åtog de sig att över tid spåra viktiga aktiviteter och införliva data i ett simuleringsverktyg som de höll på att utveckla, så att de kunde jämföra de två uppsättningarna resultat

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Approved	Examiner	Supervisor
2024-10-07	Sofia Ritzén	Susanne Nilsson
	Commissioner	Contact person
	Confidential	Confidential

Abstract

Analyzing the usage of material handling equipment (MHE) in a warehouse to increase operational efficiency and reduce costs was the main objective of this master thesis. In order to find inefficiencies and bottlenecks, the investigation examined how the present MHE was used. It also investigated the proper approach to apply for determining the size and type of future MHE. This thesis was conducted as a case study of a warehouse in Stockholm, Sweden, belonging to the Northern Europe branch of a Home Appliance Manufacturing Company (HAMC).

The research was based on the analysis of the types, quantities, and use of MHE used by the HAMC, along with an analysis of problems that employees experienced. A literature review was conducted to get a deeper understanding of the different methods used for selecting MHE and how these problems are resolved to improve operational efficiency. During this literature review, four distinct approaches to MHE selection and material handling system design were identified: conventional, analytical, knowledge-based, and hybrid. The literature review also helped in determining which key performance indicators should be tracked and how to compensate for their absence.

Modifications were made to the future MHE's composition based on efficiency criteria to enhance their performance and overall efficiency, reduce operating, maintenance, and leasing costs, and streamline operations. Key modifications included equipment elimination, system upgrades, safety protocols, parameter measurement, utilization analysis, and an MHE allocation tool. Employee participation in the research and the design of the new MHE system was crucial in identifying the real operational problems faced daily in the warehouse. By involving employees in the design process, the new system is tailored to their specific tasks and expectations, ensuring that it will be more efficient. After confirming the research and methodology, the project sponsors were pleased with the outcomes for the new material handling system's composition. In addition, they committed to track important activities over time and incorporate the data into a simulation tool they were developing so they could compare the two sets of findings.

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NOMENCLATURE

This section explains acronyms and terms used in the thesis.

Abbreviations

AGV	Automated Guided Vehicle
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
AS/RS	Automated Storage & Retrieval System
DMP	Decision-Making Process
DSS	Decision Support System
ERP	Enterprise Resource Planning
HAMC	Home Appliance Manufacturing Company
JIT	Just-in-Time
MC	Manufacturing Companies
MCDM	Multi Criteria Decision Making
MDA	Major Domestic Appliances
MHA	Material Handling Analysis
MHE	Material Handling Equipment
MHES	Material Handling Equipment Selection
MHS	Material Handling System
MHU	Material Handling Utilization
MSS	Mobile Rack Storage System
SDA	Small Domestic Appliances
SMAA	Stochastic Multicriteria Acceptability Analysis
TPS	Toyota Production System
VIB	Vendor Item Barcode
VSM	Value Stream Map
WMS	Warehouse Managing System
3PLC	Third-Party Logistics Company
3PMHES	Third-Party Materia Handling Equipment Supplier

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1 INTRODUCTION

The efficiency of warehouse logistic operations for manufacturing companies (MC) relies on the utilization of material handling equipment (MHE) for loading, unloading, moving and storing items (Gonzalez-Aleu et al., 2022; Tang et al., 2023). Choosing the right MHE is essential for increasing productivity and reducing unnecessary investment and operating costs (Hassan, 2010; Tuzkaya et al., 2010; Ulutaş et al., 2023).

Over the last years, selecting MHE has become more and more complex, mostly due to factors like shorter product life cycles, higher product quality and wider range of product choices (Tuzkaya et al., 2010).

Traditional methods, especially analytical models and expert systems, have been extensively used for MHE selection. However, these methods have limitations in dealing with all the different objectives and constrains involved in the decision-making process (DMP) (Hassan, 2010; Ulutaş et al., 2023). When it comes to managing the complex tasks of warehouses, it is very important to follow a holistic approach for material handling analysis (MHA) that goes beyond ordinary numerical models; an approach that examines various operational parameters (Tuzkaya et al., 2010; Ulutaş et al., 2023). Even though analytical models are able to offer useful insights, they often miss the complex connection of factors like operational goals, environmental impacts and user-specific requirements (Hassan, 2010).

In order to explore methods for re-designing the MHS for increased efficiency, this study primarily aims to develop a technique for analyzing material handling utilization (MHU), volume volatility, and user needs in warehouses. A Home Appliance Manufacturing Company (HAMC) that has recently adopted and implemented Lean Principles to improve its warehouse operations forms the basis of the study.

1.1 Purpose and research question

HAMC launched a new initiative in order to streamline operations, and boost the MHE's efficiency. The main objective of the project was to evaluate the current utilization of MHE and to determine the types, quantities, and specifications of the new MHE in order to enhance the efficiency of warehouse operations. Given the time constraints impacting decision-making and project scheduling, it is crucial to finalize MHE needs and place orders promptly to accommodate suppliers' extended lead times. This challenge is not unique to HAMC but is faced by many manufacturing companies, where extended lead times and complexities in timely procurement can significantly affect operational efficiency and supply chain management across various industries (Gonzalez-Aleu et al., 2022 and Tang et al., 2023). Preserving operational efficiency and competitiveness in the fast-paced manufacturing sector requires effective management of MHE procurement. Making the most of warehouse operations and minimising downtime requires the capacity to foresee and proactively handle procurement issues.

This study seeks to present a thorough technique for assessing and developing MHE systems, providing useful insights that may be applied to many manufacturing organisations dealing with similar limitations. Through the resolution of these crucial concerns, the study adds to the larger area of supply chain optimisation and material handling by providing answers that can improve strategic planning and operational efficiency in many industrial settings.

The following research question is investigated in this study to accomplish this objective and overcome the obstacles: "How can the analysis of a company's current Material Handling Equipment (MHE) utilization be designed to reveal operational inefficiencies and challenges, and

how can these insights be leveraged to develop strategic recommendations that enhance the performance of its logistics operations?".

In addressing this question, the study will focus on how existing MHE utilization records, volume fluctuation analysis, and real-time task observations and interviews can be utilized in the design of a new material handling system to meet efficiency goals. This integrated approach will guide the development of a robust framework for optimizing MHE systems, ensuring they are adaptable and efficient in managing varying operational demands.

1.2 Potential benefits

This study analyses HAMC's case study in detail to provide light on how to effectively modify MHE to fit changing product demands and dynamic warehousing settings. It lays out a realistic plan for fixing operational inefficiencies by carefully selecting and configuring up MHE.

Furthermore, by showing how certain features and configurations can address common problems and boost overall performance, the study contributes to the discussion of best practices for MHE selection. This study demonstrates the significance of ergonomic and safety issues in MHE design, provides a framework for similar future endeavours, and highlights the need to incorporate these factors for MHEs to function at their best.

The study also highlights how important it is for MHS to follow all regulations and safety protocols. Organisations may set an example for ethical material handling practices and stay in compliance with changing requirements by implementing cutting-edge safety features and technologies to reduce the likelihood of accidents and noncompliance. These steps not only improve operational safety, but they also show that the company cares for its employees as a whole, which leads to a more positive and secure work environment for everyone.

The new Material Handling System (MHS) will have substantial long-term benefits for HAMC, in addition to these more general contributions. A number of anticipated benefits are likely to have a favourable effect on HAMC's bottom line, including increased overall productivity, decreased energy usage, and decreased maintenance and leasing expenses. Optimisation of efficiency is just one goal of these operational enhancements, which also aim to make logistics operations more sustainable and cost-effective.

In summary, this research the results from this study is useful for HAMC and makes a dent in the literature on material handling systems. Anyone working to improve the adaptability and efficacy of MHE may find the results a valuable resource, since it offers both theoretical advances and practical insights that can guide future studies and industrial practices.

1.3 Case study delimitations

The research focused exclusively on operations within one company; HAMC's Northern Europe warehouse in Stockholm. Further, it was conducted within a period of eight weeks, with equal time spent on observation and analysis.

All attention was directed on 3PMHES's rented MHE, which includes industrial vehicles such as high reach, pallet, order picking, and counterbalance trucks. Data restrictions on energy consumption rates and time metrics including loading, put-away, picking, packing, stacking, and replenishment meant that other variables had to be prioritised in the study. In particular, priority placed on total daily volume demands, while product variations considered only in terms of MHE type. Since useful time metrics were not available, simulation of the engineered model was not possible, and no conclusions can hence be made regarding the optimal way to determine the size and the composition of a MHS.

In order to correctly reflect the changing volume demands, the gathering of data took into account more than one time period. After careful observation, other issue areas were found; nevertheless, the scope was ultimately reduced to a thorough evaluation of the utilisation of industrial vehicle fleets.

Every single person in the warehouse, not just the managers and team leaders, had to contribute since observation and interviews considered more important and significant due to the very basic data analysis used to identify HAMC's real operating needs in MHE.

1.4 Disposition

The structure of this master thesis is organised as follows. Chapter 2 outlines the methodology employed when conducting the research. Chapter 3 presents the theoretical framework used to guide the improvement work. Chapter 4 presents the case company. Chapter 5 provides empirical findings, including identified problems through interviews and observations, volume fluctuations and MHE underutilization problems raised by data analysis. Chapter 6 focuses on the approach used to review the empirical findings and the creation of a model for analysing future needs. Chapter 7 discusses the fulfilment of the objectives, as well as the importance of the alternative approach when analysing future needs. Chapter 8 summarizes the results of the research in order to answer the research question. Lastly, Chapter 9 reflects on findings, challenges, and obstacles that might have affected the results of the research.

2 METHODOLOGY

This study was designed to improve material handling systems (MHS) by analyzing typical inefficiencies in warehouse operations, with an emphasis on the routine work that workers do. The aim was to create solutions that can help many different people that have a hand in material handling. Through the utilization of a case study approach, valuable lessons were derived about the identification and remediation of these inefficiencies, which can be applied to comparable scenarios in different sectors.

Therefore, in order to provide a thorough analysis of the current MHE utilization, the study utilized both quantitative and qualitative methods. The necessity to record both qualitative and quantitative user experiences in addition to objective performance data motivated the choice to combine the two approaches. The quantitative methods were employed to measure operational hours and volume, which provided precise numerical insights into the performance and utilization of MHE (Creswell, 2014). In contrast, the qualitative methods delved into employees' verbal and nonverbal communication, offering deeper understanding of their perspectives, experiences, and the external factors impacting the effectiveness of MHE (Bryman, 2016).

Using a mixed-methods approach allows for a more complete understanding of material handling systems since quantitative data is strong and qualitative insights are rich (Creswell & Plano Clark, 2018). By taking a broader view, we can better comprehend the problems and create solutions that are both successful and centered around the needs of the users (Tashakkori & Teddlie, 2010).

The new MHS could not have been built, tested, or validated without the staff's active participation. The new system was more likely to be adopted and used successfully since their direct involvement not only made sure it solved the practical problems, they had with the old MHE, but it also enhanced the possibility of that happening. In order to find inefficiencies and make sure the revamped MHS adequately suited staff demands, feedback was crucial (Robson and McCartan, 2016). The employees' demand for change, stemming from underutilization of current MHE, inefficient operations, and safety concerns, drove their active engagement in the study process. A crucial component in the research's success was the useful insights supplied by their interaction (Glesne, 2016).

2.1 Research design

The working approach followed in this project was divided into six sections, as briefly illustrated in figure 1. Initially, a short pre-study was performed to understand and clarify the goals, objectives and scope of the project. At this point, the key stakeholders were also identified, and communication channels established. A first literature review followed in order to allocate the scientific findings, tools and methods used to approach such problems and also to figure a possibility for new research contribution. This first literature review helped in the formulation of the research question for the thesis.

An empirical study with huge amount of data collection followed, which included interviews, observations and existing documentations reviews. In order to reveal problems and areas for improvement a comprehensive analysis took place where collected data cleared, categorized and verified.

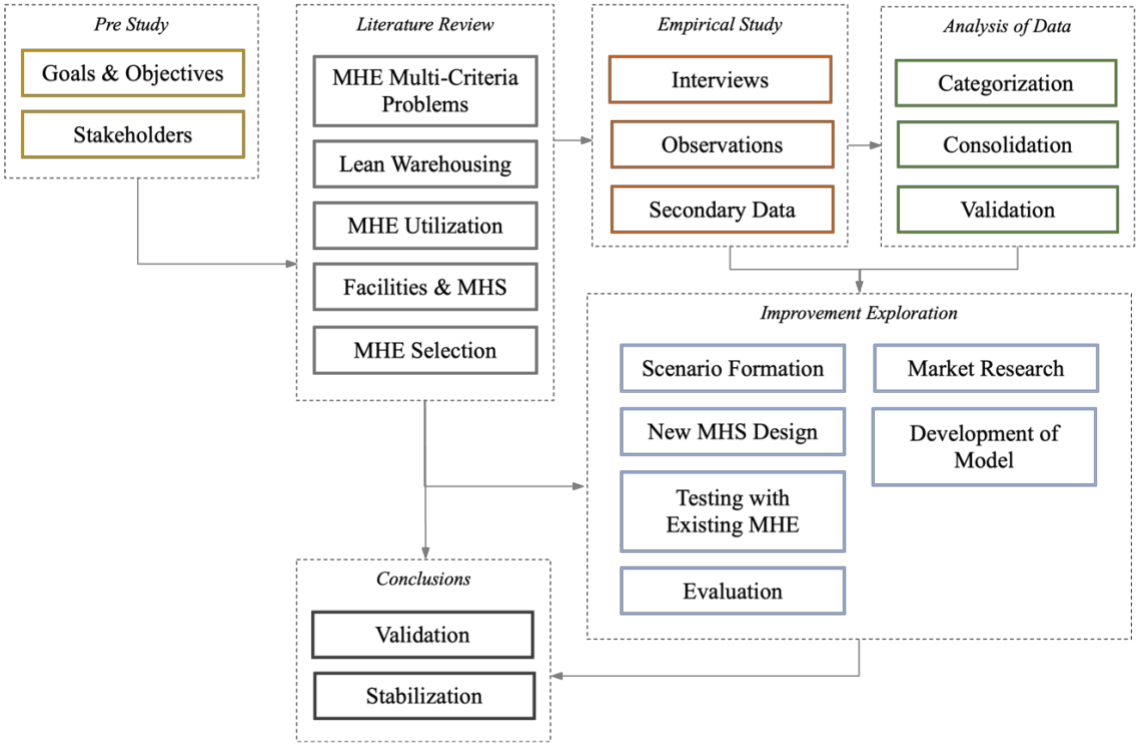


Figure 1. The working approach used in the research.

The findings from the empirical study guided a second literature review, in order to find inspiration on how the new MHS could be designed to be more efficient. At that point, the literature study extended to areas such as lean warehousing and MHE for lean automations, since HAMC's operations are lean based, with an aim to find best practices in the design of a lean MHS.

After the second literature review, an extensive scenario analysis took place in order to address the future volume demands for the new MHS. Then, an iterative design process started, where the existing MHS was improved through a four steps design process; market research, design, testing and evaluation. In the market research step, new technologies, tools and equipment were identified, in the design step the market research findings evaluated and a selection was made of choices that solves the identified problems; this by analyzing the problems according to the literature found about lean warehousing. The newly designed MHS was also compared to the best practice found in warehouse MHS design and MHE selection literature, as well as compared to the existing MHS in HAMC's warehouse in Stockholm and other sites abroad, in order to find additional inspiration on how to improve the efficiency of the system.

The suggested MHS was then discussed with the employees, adjustments on current MHS were made to make it look like the suggested MHS in order to test it and see how well it fits the volume demands and the needs and wishes of the employees. The findings were then evaluated and used as an input for the redesign process where the MHS was further improved to achieve maximum operating efficiency. The design process repeated a few times until a satisfactory MHS was achieved. The working process, the findings from the analysis and the designed MHS were then validated from the sponsors of the project, ensuring their accuracy and effectiveness; which aimed to assess the stability of the project outcomes, confirming that the results were reliable across different perspectives and scenarios.

2.2 Pre-study

At the beginning of the project a pre-study was performed to understand and clarify the MHS of a HAMC's warehouse in order to define the problematic area and the scope of the research. A

primarily data collection from the project sponsors and warehouse managers, and a secondary data collection from major MHE suppliers within the field of home appliances took place, in order to understand the goals and the objectives of the project and to identify different stakeholders and establish communication channels with them.

The pre-study resulted in the development of an approach strategy where many different aspects identified and set foundations of the literature review and the further investigation of the design of an efficient MHS to be implemented in the warehouse of a home appliance manufacturing company who applies lean warehousing principles.

2.3 Literature review

A thorough literature review of books, internal documents, journal articles, and literature from closely related fields was conducted to ensure careful examination and review of both supporting and contradicting findings to empirical data.

The literature reviews were conducted twice, once at the beginning of the project, and again before the improvement work phase, using a two-step process. Initially, the literature was scanned to assess its relevance with the field and obtain a basic understanding. If considered relevant, a more thorough examination was conducted to gain a deeper understanding and assess its validity.

The initial literature review aimed to discover the main focal of previous research in the domain of MHS design, and to ensure that the project's topic will add to this knowledge field. Additional study was conducted on multi-criteria MHE selection problems to explore possible methods for improving MHE selection processes. The identified methods were then evaluated for their applicability to the project's objectives.

The second literature review focused on MHS specifically designed for lean warehousing. The objective was to find key aspects to consider while designing a MHS that follows lean principles. These aspects were then used to establish criteria for guiding the design of the MHS. Furthermore, this review aimed to collect insights and recommendations that may be used to improve operational efficiency by enhancing the MHE selection process. The second literature review was mainly conducted before and during the improvement work stage and was guided by the findings of the empirical study analysis. This second literature study was designed to be continuous, in order to keep exploring methods and solutions based on emerging problems during the project.

The trustworthiness and authenticity of the literature, was secured by exclusively relying on published articles and books. If any information was considered questionable, efforts were made to verify its accuracy by exploring additional sources. The publication date of each source was verified to ensure the relevance and currency of its information. Preference was given to primary sources, and whenever an article referenced another, the original source was always investigated. In cases where access to the primary source could not be obtained, secondary sources were cautiously used as alternatives.

2.4 Empirical data collection

The empirical research consisted of both quantitative and qualitative data collection strategies. The investigation was conducted in order to understand the HAMC's needs and preferences regarding the selection of MHE, whereby the outcome was later used in further analysis. The methods used for data collections were interviews, observations and secondary data.

Interviews

Interviews were used extensively in the research as a method of empirical data collection as shown in table 1. Unstructured and semi-structured interviews were used to gain an in-depth understanding of the daily MHE operations, the expectations and limitations of the new MHS, as well as to gather employee's experience and suggestions for improvements of the MHE (Kvale et all, 2009). Examples of the questions used in the interviews can be found in Appendix 1.

HAMC Employee	Duration of the Interview	Follow up Meeting
Head of Logistics	60 minutes	
Logistics Manager	75 minutes	2 (20 min/interview)
Warehouse Operations Manager	60 minutes	7 (10 min/interview)
Transport Operations Manager	60 minutes	2 (20 min/interview)
Quality Operations Manager	30 minutes	
IT Manager	30 minutes	
Operations Safety Manager	60 minutes	3 (20 minutes)
Warehouse Operations Manager, Transport Operations Manager, Operations Safety Manager	60 minutes	2 (20 minutes)
Warehouse Operations Manager, Operations Safety Manager	30 minutes	5 (15 minutes)
Warehouse Operations - Team Leader 1	60 minutes	2 (30 min/interview)
Warehouse Operations - Team Leader 2	40 minutes	1 (10 minutes)
Warehouse Operations – Operator 1	20 minutes	1 (10 minutes)
Warehouse Operations – Operator 2	30 minutes	1 (30 minutes)
Warehouse Operations – Operator 3	20 minutes	1 (10 minutes)
Warehouse Operations – Packer 1	20 minutes	1 (10 minutes)
Warehouse Operations – Packer 2	20 minutes	1 (10 minutes)
Warehouse Operations – Picker 1	20 minutes	1 (20 minutes)

Warehouse Operations – Picker 2	20 minutes	1 (20 minutes)
Warehouse Operations – Picker 3	20 minutes	1 (20 minutes)
Warehouse Operations – Team Leader 1, Operator 1, 2, 3	60 minutes	2 (20 minutes)
Warehouse Operations – Team Leader 1, Picker 1, 2, 3	40 minutes	1 (20 minutes)
Warehouse Operations – Operator 1, 2, 3	40 minutes	3 (20 minutes)
Warehouse Operations – Picker 1, 2, 3	60 minutes	3 (30 minutes)
Warehouse Operations – Packer 1, 2	30 minutes	1 (20 minutes)

Table 1. Overview of interviews and follow up meetings.

An interview described as an everyday conversation that has a structure and purpose; to get as much detailed information as possible by the interviewee (Kvale et all, 2009). The interviews phase started with unstructured interviews in order to define the objectives and directions of the research (Westlander, 2000). At that time, employees and managers were uninfluenced from previous conversations and their true view was investigated. The aim of unstructured interviews was to identify the problem areas that the employees and the managers decided that wanted to bring up and thought were important for them and significant for the research.

Semi-structured interviews followed, that allowed to focus towards the goal of the research while still keeping an open approach to what questions could lead to that goal. This method was chosen to ensure a comprehensive exploration of the research theme while adapting to the flow of conversations (Johnson & Turner, 2003) Most conversations at that stage followed a pre-determined, questions-guide approach with a list of themes needed to be covered. However, the employees and the managers were those who guides the flow of the conversations with follow-up questions to be added to go deeper into areas that the interviewees mentioned.

Since interviews were used daily as a method of data collection, the questions asked were carefully considered in order to avoid leading the conversation to false input. During the interviews notes were taken extensively and at the same time audio and video recordings used as a back-up for later analysis. This dual approach ensured the accuracy and richness of the data collected.

Observations

Observation was used as a data collection method throughout the project, with the participants observed in their natural environment. It was crucial to observe all workstations and every step of the warehouse processes because usually people act normal when they don't know they are being observed, especially at the beginning of the project. Observations aimed to validate interview findings and uncover previously unidentified problems, and additional data (Jason & Turner, 2003).

The observation period lasted for 4 weeks and took place in the HAMC's warehouse, with daily observations of employees using the current MHE. Observations were also used during the testing phase, where employees attempted to adjust current MHE to simulate operations expected the newly designed MHS. This was done to assess how the new MHS could facilitate operating processes more efficiently.

When conducting the observations, two different approaches were used; the complete participant and the observer as participant approaches (Lewis et al., 2009; Johnson and Turner, 2003). HAMC operates in two shifts, one morning and one evening shift. These shifts overlap only for a few minutes and employees from the morning shift don't have the time to extensively exchange information with employees from the evening shift.

At the beginning of the project, warehouse employees were not aware of the project, and the complete participant approach was employed for a couple of days. Employees from both shifts were not informed by their team leaders about the presence and the purpose of a new person in the warehouse, as they were also unaware. While applying the complete participant approach, the observer attempted to fully integrate with the warehouse employees and engage in their activities without informing them that they are being observed, in order to collect genuine qualitative data.

After applying the complete participant approach for two days, the new person in the warehouse was known and recognized by most of the warehouse employees, and everyone became aware of the project, that person's purpose and that their actions and work were being observed. Consequently, the observer as participant approach had to be adopted.

While applying the new approach, the observer aimed to remain as neutral as possible; attending warehouse operations without actively participating in them and minimized interaction with the employees to avoid disrupting their routine and collect accurate data. This approach was very helpful during the testing the newly designed MHS, ensuring that the opinion of the employees was not affected by external factors.

During the all observations, field notes were taken extensively and at the same time audio and video recordings used as a back-up for later analysis.

Secondary data

At the beginning of the project, stakeholders were identified, clarifying their roles and the information and type of data they could share, a preliminary list of relevant documents was developed. The list continuously updated throughout the project, ensuring that relevant information was systematically collected.

The secondary data collection process involved extensive exploration of HAMC's SharePoint platform, while a tremendous amount of time spent to navigate through various folders and files. At that point stakeholders from different departments contributed by providing valuable information and recommendations for documents that could offer insights and were not considered initially. Their contribution broadened the scope of data collections, the range of perspectives and sources for the project.

The documents that were utilized as secondary data were generated prior to the initiation of the project by HAMC's different departments for diverse purposes and categorized in folders on the department's SharePoint platform. These documents included warehouse layouts, operational process maps, volume balancing optimization proposals, HAMC's product portfolio, list of approved MHE, overview of MHS compositions across other HAMC's warehouses, volume traffic records, and MHE leasing costs, among others.

3PMHES's Connect platform was another source of secondary data; this online tool, offered by 3PMHES as a service of MHE management, provided additional relevant information about the

utilization of the current MHE. Thought this combined data collection approach, a rich secondary data base was developed, which helped to better understand processes and the problematic area.

2.5 Analysis of data

The utilization of both qualitative and quantitative data collection strategies resulted in a large amount of data for analysis, enabling for a cross-validation of the results. The empirical findings were analyzed in four stages; categorization, clearance and consolidation, and validation, aimed at assessing the constrains of the current MHE's operational capabilities, and the actual operational dynamics within HAMC's warehouse.

During the first stage of the data analysis, exhaustive examination of documentation from interviews, observations, and secondary data, was conducted to identify key insights. Findings were systematically summarized and organized into groups of broader problem areas; a long process with multiple revisions until distinct problem areas emerged clearly.

Underutilization of equipment and inefficient procedures were the most notable operational inefficiencies. According to the results, inefficient procedures and antiquated hardware were the root causes of the problems. In particular, these delays were caused by manual handling duties, and it was clear that scheduling and task allocation were inefficient.

Another big issue was the equipment, which was plagued by insufficient capacity and antiquated technologies. The data demonstrated that material handling activities were severely disrupted due to older equipment failing to fulfil the needs of higher order quantities.

Inadequate training programs, high turnover rates, and skill mismatches were recognised as important factors contributing to the training and staffing challenges that were also apparent. Staff skill gaps were a result of insufficient training, which, when coupled with high turnover rates, further disrupted the workforce and reduced operational efficiency. Poor performance in material handling tasks was associated with a lack of training and a mismatch between skills.

Outdated safety procedures and an absence of safety training were highlighted as major issues with compliance and safety. According to the results, these antiquated processes and a lack of staff knowledge of safety practices were associated with safety occurrences.

Poor interdepartmental communication, disjointed task execution, and insufficient feedback mechanisms were the main communication and coordination concerns impacting the material handling procedures. Evidence from material handling operations shows that inefficient communication and feedback channels lead to frequent misunderstandings and delays.

The purpose of categorizing the findings in this manner, was to highlight broader problem areas within the current MHS, and to make it easier to identify trends across problem areas. This approach also simplified secondary data cleaning and consolidation processes. The analysis of the data contributed to the formulation of initial requirements for the new MHS, and provided strategic guidance on specific topics for the upcoming literature review; enabling further exploration of solutions and approaches to solve identified problems.

2.6 Improvement exploration

The majority of the project focused on the engineering phase, aimed at improving the MHS by selecting MHE to increase operational efficiency within HAMC's warehouse. Throughout this phase warehouse employees and key department managers engaged in an iterative process to ensure that employee and managerial objectives will be met. Involving warehouse employees was intentional, not only to gather input improving operational efficiency but also to gain their support for the new MHS, whose design was "controlled by them".

The iterative development methodology, ensured that warehouse employee's considerations, that were continuously emerged across several iterations, were integrated into the final MHS design. This methodology was chosen for its ability to rapidly develop an initial design for testing and evaluation by warehouse employees, providing valuable feedback for further refinement.

Following each evaluation, MHS requirements were updated, and a new iteration began with the development of an improved design, which in turn was evaluated by the warehouse employees. This iterative process of designing and testing continued until the optimal MHS was achieved, with the number of iterations determined by feedback from warehouse employees.

During the testing phase, warehouse employees were asked to adopting the current MHS to simulate the new design, that happened in several iterations which allowed the employees to investigate by themselves the efficiency of the proposed solution. While improvements were made to both the overall design of the MHS and to specific MHE, the simulation focused on individual workstations due to the inability to stop warehouse operations for a full-scale simulation. Following the adaptation of the current MHS, employees were asked to evaluate the new system and give their feedback and suggestions for further improvements. Their feedback and suggestions were then analyzed and evaluated, leading on additional requirements for the next design iteration. The results were discussed and presented to warehouse employees who participated in the process to validate the outcomes.

Upon completion of the new MHS design and investigation of MHE suppliers and their offerings, the study was handed over to HAMC in order to assess the quality of the research and its outcomes and validated the proposed design and selections. All records, audio and visual, and documentation of interviews, observations and secondary data were uploaded on SharePoint for examination in order to increase the reliability of the outcomes. Subsequently, HAMC validated the quality of the research and its outcomes, though due to the long waiting time for the new equipment to be delivered, implementation of the new MHS wasn't possible within the timeframe of this study.

3 THEORETICAL FRAMEWORK

The theoretical framework utilized in this study focuses on MHS design and MHE selection processes to address the research question: "How can a company's current Material Handling Equipment (MHE) utilization be analyzed to identify operational inefficiencies and challenges, in order to develop strategic recommendations that enhance the efficiency and effectiveness of its logistics operations?" This section focuses on the technical aspects of system design and the selection of appropriate equipment to support warehousing principles and operational goals. It draws on frameworks and methodologies to ensure that the MHE selection process is robust and aligned with the dynamic needs of the warehouse environment (Hassan, 2010; Tuzkaya et al., 2010).

In the context of the case company HAMC, the application of lean methodologies by HAMC's managers has already led to significant reductions in operational costs by identifying and eliminating machinery that was later deemed unnecessary. This approach not only streamlined operations but also enhanced the adaptability of the MHE to changing warehouse requirements (Gonzalez-Aleu et al., 2022; Hines et al., 2004). Examining HAMC's case sheds light on effective strategies for adapting MHE to accommodate dynamic warehouse environments and fluctuating product demands. This approach offers a practical framework for addressing operational inefficiencies through the strategic selection and configuration of MHE (Ulutaş et al., 2023).

In summary, while the primary focus of this study is on the design and selection processes of MHS and MHE to achieve operational efficiency, the involvement of employees as stakeholders in the design and configuration process serves as an additional layer of improvement. This approach highlights the necessity of incorporating human-centered research methodologies, which challenge the customary reliance on quantitative data alone, and ensures that the new MHE system is tailored to the real-world challenges faced by those who interact with it daily (Kotter, 1996; Maguire, 2001; Sanders & Stappers, 2008).

This comprehensive approach is essential for developing a robust framework for optimizing MHE systems, ensuring they are adaptable and efficient in managing varying operational demands. The literature sections provide the necessary background and justification for this integrated approach, highlighting its relevance and applicability in achieving the research objectives.

3.1 Warehouse operations

Warehouses play a vital role in proper functioning of the supply chain by providing temporary storage for goods and facilitating inventory distribution, sorting, and cross-docking, leading to increased operational efficiency and improved customer service (Sinha and Muralidhar, 2022). Receiving, inspecting, repackaging, putting away, storing, order picking, sorting, packing, and shipping are all part of warehouse operations (Tompkins et al., 2010)(Fig. 2). Understanding these operations and their associated issues (as detailed in Figure 2 and Table 2) provides the foundation for identifying inefficiencies and areas for improvement in the MHS. This literature helps identify specific operational inefficiencies and errors that the new MHS should address to enhance overall warehouse performance.

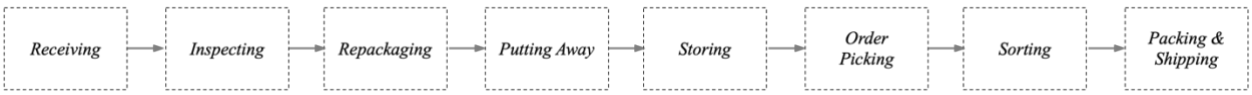


Figure 2. Warehouse Operations. Source: Tompkins et al.(2010)

i. Receiving: It is the process of receiving products once an order has been made. At this stage, warehouse employees verify that the products' quantity and quality meet the order prior to distribution for storage (Sinha and Muralidhar, 2022).

ii. Inspecting: Both the quality and quantity of products received prior to distribution for storage are checked via the process of inspection and quality control (Sinha and Muralidhar, 2022).

iii. Repackaging: If the quantity of materials received is more than the storage cube need or the product does not have any markings that can be recognized by either people or system, repackaging will be performed (Sinha and Muralidhar, 2022).

iv. Putting away: It is the process of transferring goods from unloading docks to storage facilities. At this stage, the primary tasks are handling and placing materials (Sinha and Muralidhar, 2022).

v. Storing: Products affect storing. The required storage space depends on product's dimensions, volume and characteristics (Sinha and Muralidhar, 2022).

vi. Order Picking: The process begins when the company receives an order from a client. Part of the process is the selection of pallets or smaller quantities from the bulk area storage or the pick area storage. (Sinha and Muralidhar, 2022)

vii. Sorting: Sorting is the process of arranging batch picks into customers' orders. At this point, orders that include more than one items are facilitated with picks (Sinha and Muralidhar, 2022).

viii. Packing and Shipping: This is the process of verifying and putting completed orders into shipping . Later on, shipping documents are prepared and the goods are loaded into the transportation means (Sinha and Muralidhar, 2022).

Various activities within the warehouse operations provide enough opportunity for errors (Tompkins et al., 2010). Table 1 contains a comprehensive list of activity issues.

Warehouse process	Problem description
Receiving	Carton damage
	Mismatch b/w items information and freight bill
Inspecting	Quality defect
	Less quantity delivered
	Wrong item delivered in cartons
	Missing items
Repackaging	Item break due to poor handling
	Wrong labelling on cartons
Putting Away	Item break during movement to storage location
	Item damage due to poor handling
Storing	Item put in wrong shelf
	Storage space not optimally utilized
	Excess stocking
	Product damage during storage
Order Picking	Wrong item picked
	Delay in identifying item location
	Wrong quantity picked
Sorting	Accumulation of wrongly picked orders
	Excess quantity sorted
	Less quantity sorted
Packing and Shipping	Wrong item packed / shipped
	Product damage due to poor material handling
	Error in shipping document preparation
	Error in packing and labelling

Table 2. Activity issues in warehouse operations. Source: Tompkins et al. (2010)

3.2 Material handling systems design

Warehousing serves an essential role of 'buffering', which tries to maximize storage utilization and promote efficient material management (Frazelle, 2002). Material handling is highlighted as a waste in a lean warehouse setting and should be considered for cost reduction (Kilic et al., 2012).

Appropriate storage and handling equipment selection during warehouse design is critical since it increases material flow, storage utilization, and productivity; a selection that should be made with careful consideration since affects operating costs and performance over the warehouse's lifecycle (Shah and Khanzode, 2015).

Material handling operations use 80% to 95% of the time required to process orders and account for 20% to 50% of the overall operating expenditures (Tompkins et al., 2010). This indicates how efficient handling devices and storage systems may boost productivity, decrease reaction time, and maximize space and equipment utilization, all while leading to substantial cost savings.

An approach to design a material handling system is to examine and evaluate different layouts and material handling alternatives (Heragu et al., 2010). Decisions on process flow, warehousing system selection, and other operational rules are part of the strategic design for storage and handling systems (Rouwenhorst et al., 2000). The process flow specifies the sequence of steps that materials undergo throughout operations including receiving, storing, picking, and shipping. Capacity needs, budget, quantity, and application process type (e.g., storage, handling, picking, etc.) are the criteria used to assess material handling systems (Gu et al., 2010). Important performance indicators influencing warehouse metrics are MHS cost, quality, time, and productivity; which should be carefully considered at the design stage since they impact the warehouse's operational cost and performance (Rushton et al., 2006, Baker and Canessa, 2009; Gu et al., 2007; 2010; Staudt et al., 2015).

Structure, layout, sizing-dimensioning, operational strategy, and equipment selection are the five categories of design difficulties and associated considerations (Shah and Khanzode, 2015). It is important that the handling devices can adapt to the different kind of loads needed for various activities, including picking, storage, receiving, and dispatching (Rushton et al 06). Storage systems and handling devices impact each other's performance and must be chosen at the design stage, since changing them later is difficult (Shah and Khanzode, 2015). There are two types of storage handling systems: pallet and non-pallet as presented in Figure 3.

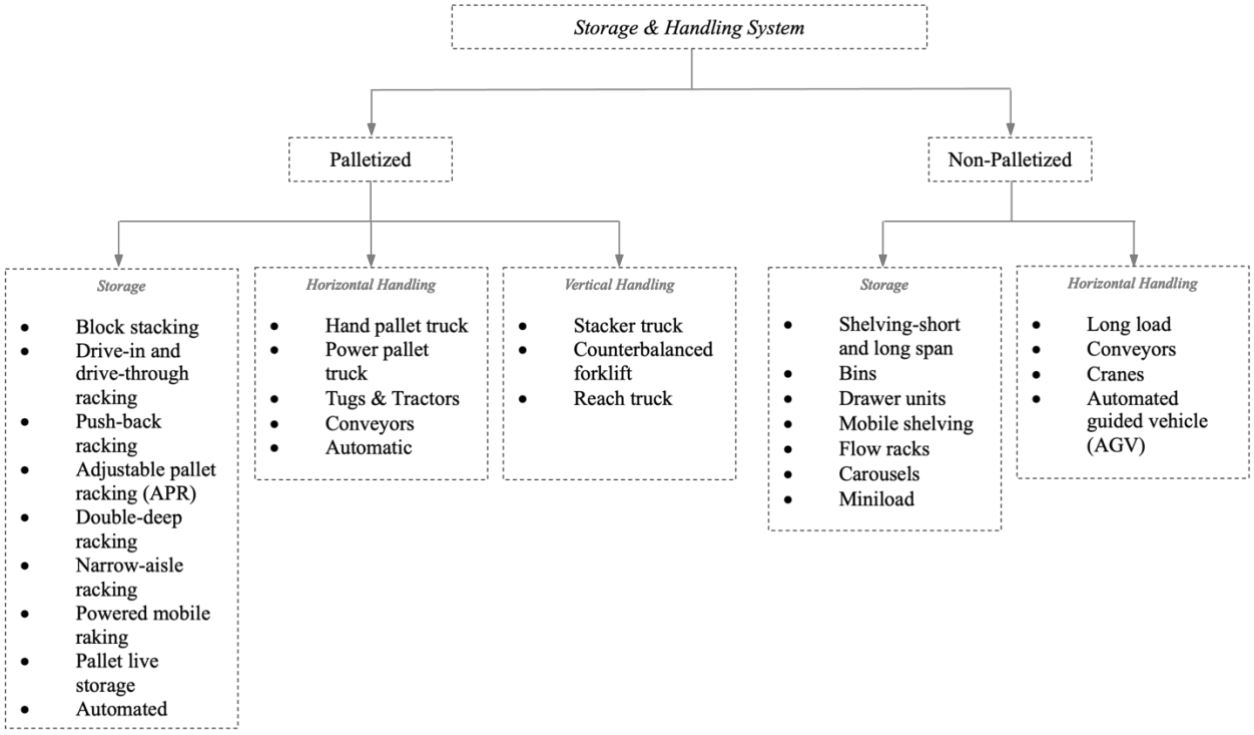


Figure 3. Taxonomy of storage and handling systems. Source: Shah and Khanzode, 2015.

The wooden pallet is the most common used unit load in warehouses and allows standard storage and handling equipment to be utilized, regardless of the kind of products. Several aspects influence equipment design, including throughput levels, inventory holdings, and supply chain needs (Rushton et al, 2006). Some products, however, have special features such as being too tiny, too big, too long, or requiring lifting from the top, making them incompatible with the palletized system (Shah and Khanzode, 2015).

There are various storage solutions available for both palletized and non-palletized systems and simulation is often used at the research level to aid decision makers in making design choices. The majority of simulation-based research concentrated on discussing individual sub-problems at the operational level, with just a few simulation studies at the design level (Rouwenhorst et al. 2000; Gu et al. 2010). Options for palletized systems range from high density and restricted access to minimal storage and complete accessibility, such as block stacking (no racking), racks (physical racks), and automated storage and retrieval system (AS/RS) (Shah and Khanzode, 2015).

Block stacking involves placing pallets directly on the floor and building them up in a stack with no storage equipment support and very little space. The use of counterbalanced forklifts allows for the free removal of the top and front of each row of products. The pallet top should be flat enough to serve as a secure basis for the following one and capable of supporting overlaid weight. To minimize duplicate handling, each row contains the same SKU (Shah and Khanzode, 2015). The benefits include efficient space utilization, layout flexibility, and fast inventory access with high throughput (Shah and Khanzode, 2015). However, there is a possibility of storage wastes; selecting lane depths is a strategic choice that balances the tradeoffs between space utilization and convenience of storage/retrieval operations (Gu et al., 2010).

Rack storage systems are functionally similar but might vary in design to meet specific needs. Various types of systems are available based on storage and retrieval conveniences, costs, and space utilization, including drive-in/through racking, push back, adjustable pallet, double deep, narrow aisle, pallet live, and powered mobile racks (Rushton et al., 2006). A common AS/RS setup involves high-bay pallet racking, with a stacker crane moving along the aisles of the racks to store and retrieve pallets. A storage media, storage-retrieval machinery, feed-in/out equipment, and controlling software comprise this system (Shah and Khanzode, 2015).

Shelf units, tote-bins, drawers, mobile shelves, carousels, and miniloads are all options for storing small items without using pallets; the only things that set them apart are their size, material, and operational characteristics (Heragu et al 2010). A storage rack system may be either stationary or movable. The movable rack storage system provides for the storage of objects or products that may subsequently be moved to different sites, as well as racks. Due to its higher storage utilization rate (85%) compared to a static rack system's (40%), the Mobile Rack Storage System (MSS) is also known as a High Storage Volume System (Hu et al., 2009). Low order density and deep rack rows of heavy products are better suited to this kind of system. Manual operation, huge rack row movement, restricted height, and lengthy access time are some of the disadvantages of this system (Shah and Khanzode, 2015).

Carousel is used in a part-to-pick method, and its goal is to minimize operator movement while accessing inventory. It transports goods along the required product line by storing them in shelves or that are supported by chains. Not only is it available in horizontal and vertical formats, but it also offers faster stock access and increased security (Shah and Khanzode, 2015). Miniload, also known as a AS/RS system, is a computer-controlled crane that moves down a central aisle to carry out storage and retrieval operations. It stores or retrieves goods from shelves or arranged on each side of the central aisle. It can transfer more than one carton or bin at a time to increase access speed (Shah and Khanzode, 2015).

Material handling in warehouse is known as moving goods from one location to another or from one value-added activity to another, such as receiving, shipping storing, picking, etc. (Van den berg and Zijm, 1999). Several handling devices are available in warehouses for both horizontal

and vertical movements, such as pallet moving and pallet stacking (Shah and Khanzode, 2015). Examples of common warehouse handling equipment for horizontal pallet movements are hand pallet trucks, powered pallet trucks, tugs and tractors, conveyers, automated guided vehicles (AGV); while examples of common warehouse handling equipment for vertical pallet movements are stacker trucks, counterbalance forklift trucks and reach trucks. Other equipment is also available, including narrow-aisle trucks, stacker cranes, and double-reach trucks. (ibid).

A hand pallet truck is a kind of truck that uses a simple pump action to elevate its forks, allowing the operator to manually lift a pallet off the floor and place it as needed and is commonly used for infrequent movements over short distances (ibid). Pallet trucks have similar functionalities but are powered by batteries and operated by operators (ibid). Tugs and tractors are designed as a lean handling device that is able to pull several trailers at once to shorten the distance of a normal lengthy run. Conveyers comprise a series of rollers that are angled at an angle; when a pallet is placed on the conveyer, it will move forward as a result of gravity. In the context of constantly moving items, it increases operational efficiency and decreases delivery errors (Katayama and Bennet, 1996). AGVs are driverless trucks which are computer-controlled and run entirely on electricity. They move pallets from the receiving area to the reserve area or the marshalling area, and they also interact with other handling devices. They are often used when manual handling is either time consuming or too expensive, such as when moving electronics or tiny pieces (Shah and Khanzode, 2015).

Pallets can be pilled one on the top of the other to create racks for practical storage. Stacker trucks are lightweight battery-powered trucks, available in pedestrian, stand-on, and ride-on versions, who drive their legs into the area under the bottom pallet to load or unload storage racking (Shah and Khanzode, 2015). Counterbalance forklift trucks are frequently used in outbound and outbound area. Since the forklift's cargo is carried forward on the front wheels and the truck experiences twisting actions that tends to tip it forward, a counterbalanced weight is integrated into the back of the truck (ibid). Reach trucks are used in many non-automated warehouses with confined spaces; when loading and unloading, the truck is rotated through 90 degrees so that it faces the spot where the cargo is to be placed or retrieved, with the mast to extend forward to accomplish the task (ibid).

Material handling has the potential to become a system bottleneck and cause a lot of wastes such as transportation costs, waiting times and idleness if it is poorly planned (Kilic et al., 2013). It is crucial to choose the most appropriate material handling system and put it into action effectively, in order to eliminate material handling waste, one of the seven wasters in lean warehousing, which aims to improve quality, reduce time and cost, and increase organizational effectiveness (Shah and Khanzode, 2015). Figure 4 illustrates a conceptual framework to design lean storage and handling systems. A 10% to 30% reduction in cost is predicted in case that a well-designed material handling system is implemented (Drira et al., 2007).

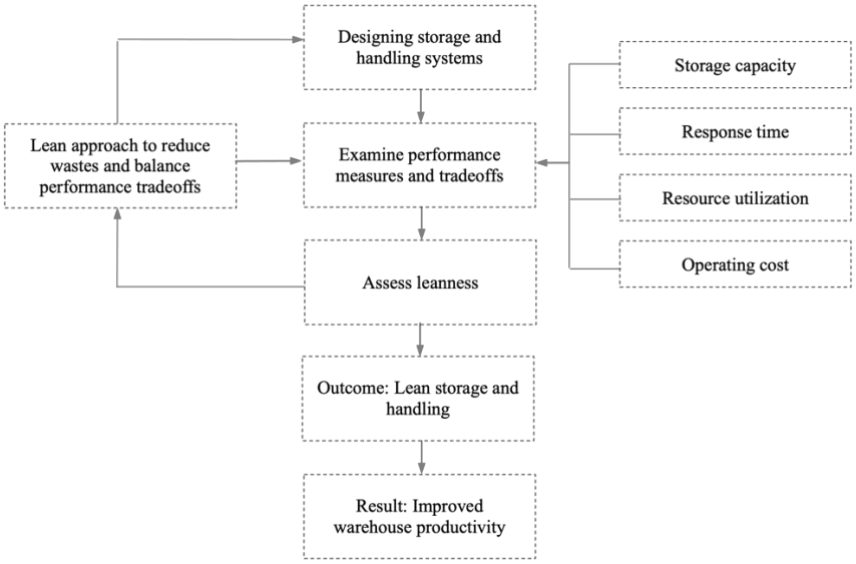


Figure 4. Conceptual framework to design lean storage and handling systems. Source: Shah and Khanzode, 2015.

3.3 Material handling equipment selection methods

Every warehouse system has unique characteristics; thus, it is essential to recognize that there is no universal solution for material handling systems; the optimal equipment may vary over time depending on how warehouse environment changes. Out of 48 logistical considerations within the warehouse, Material Handling Equipment Selection (MHES) is one of the most important (Riopel et al., 2005). There has been a total of 62 distinct approaches to MHES, either directly or indirectly addressing the subject, since the field's inception in the 1970s. These approaches have ranged from simple selection guides (Apple, 1977; Rudenko, 1971) to sophisticated decision support systems (DSS) (Garcia et al., 2022).

Selecting the best Material Handling Equipment (MHE) depends on a number of factors, and narrowing down a list of needs to a manageable number may be a time-consuming and complicated process (Saputro et al., 2015). MHE selection can be classified into three levels, depending on the focus related to MHE categories, types, and models: High, intermediate, and low level (ibid). At the high level, the primary goal is on identify which MHE category is needed, such as conveyors, cranes, industrial trucks, or positioning equipment. At the intermediate level, the objective is to select the best type of MHE within a specific category. For example, decision-makers may need to choose among various types of conveyors, such as chute, wheel, or roller conveyors. At the lowest level, the goal is to select the most suitable model of MHE within a particular type. For example, comparing several hand pallet truck models according to their features, size or other criteria (Saputro et al., 2015).

With so many potential outcomes, narrowing down MHE to a low-level problem could be challenging. The number of viable alternatives is not sufficiently reduced when the MHE selection issue is captured at a high level. Whereas selecting MHE from intermediate level reduces the possible alternatives to be chosen from 15 to 50 (Saputro et al., 2015). Within the intermediate level, two distinct categories emerge: 1. Selection within groups, with an emphasis on selecting a specific MHE type among categories among several subtypes (for instance, if the initial assessment indicates that an industrial truck is the best equipment category to carry out the material transport, the final decision will determine which kind of industrial truck is most suitable, such as an e-counterbalanced truck). 2. Selection within a specific-group, with a goal on identifying the most appropriate type of MHE given a specific category initially. For instance, since the problem description begins with the category; the desired outcome is to select the most appropriate

industrial truck type from the available alternatives (IC-counterbalanced truck, E-counterbalanced truck, narrow aisle truck and E-pallet truck) (Saputro et al., 2015).

It is important to clearly identify the pre-determined condition that relates to the problem's vital information and consideration when experienced with a situation involving MHES, as it helps in determining the most appropriate alternative (Saputro et al., 2015). To address the MHES problem in its context, the problem defined by capturing it under the given information and current conditions in the selection consideration. The problem regarding MHE or equipment selection is categorized as either multi-objective, single-objective, or multi-criteria (multi-attribute) (ibid).

In case the alternatives are evaluated using criteria, the MHE selection is characterized as a multi-criteria problem. In this context, the decision-maker's intuition, opinion, knowledge, and expertise serve as the foundation for assembling the criteria and conducting the evaluation. Such MHE selection problems evaluate alternatives based on three main criteria: material, move, and technique; each of these factors have several sub-criteria. If the alternative is selected based on more than one objective, the MHE selection is characterized as a multi-objective problem and an objective cannot be compromised in order to attain another. Objective functions and restrictions are often built from measurable information and conditions according to mathematical model rules. Selecting a MHE could involve looking for one that maximizes fitness while minimizing costs (Zhang et al., 2013) or one that maximizes machine utilization while minimizing costs (Mahdavi et al., 2011) associated with material handling, machine setup, and machine operation. Although it has some similarities with the mathematical model-based multi-objective selection problem, MHE selection is considered a single-objective problem when the objective determines the decision-making criteria (Saputro et al., 2015).

The context of MHE selection problem and the approach of solution are aligned. Different contexts call for different approaches to the same problem. When it comes to the MHES dilemma, there are four main approaches to tackle it: multi-criteria decision making (MCDM), artificial intelligent (AI), optimization, and hybrid (Saputro et al., 2015). When the problem is described as a multi-criteria selection problem, the MCDM approach may be appropriately adopted. The alternative assessment under evaluation criteria is the backbone of the MCDM approach. Some of the often-used methods included in MCDM approach are listed in table 3. Modern, sophisticated decision-making utilize fuzzy logic to evaluate and analyze data that is ambiguous, subjective, uncertain, or otherwise difficult to specify. With all the focus on fuzzy logic, it has evolved into a core component of MCDM (ibid).

Acronym	Method	Description
AHP	Analytic Hierarchy Process	Pairwise comparison procedure based on linguistic scale to compare the importance of criteria and desirability of alternatives against criteria.
ANP	Analytic Network Process	More general form of AHP. ANP structures the decision problem as a network.
ELECTRE	Elimination Et Choix Traduisant la REalité, (Elimination and Choice Expressing Reality)	Family of MCDA methods based on outranking relations between alternatives.
MAVT, MAVA	Multi-Attribute Value Theory/Analysis (including e.g. MACBETH, Simple Added Weighting)	Overall priority values of alternatives are calculated based on the objectives' weights, value-functions and performance scores of alternatives.
MAUT, MAUA	Multi-Attribute Utility Theory/Analysis	Extension of MAVT, includes probabilities and risk attitudes to form utility functions.
PROMETHEE	Preference Ranking Organisation Method for Enrichment Evaluation	Calculates positive and negative preference flows for each alternative based on the pairwise comparisons of the alternatives.
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution	Ranks alternatives using the geometric distance from the positive and negative ideal solution.

Table 3. MCDM methods. Source: Marttunen et al., 2016.

Artificial intelligent approach is known as a computerized method which consults a knowledge base to solve a selection problem and simulates the intelligence of experts (ibid). Expert systems and decision support systems are the most popular AI approaches for MHES problem solving. Due to its effectiveness as a software that captures human knowledge, expert systems are more applicable for complex systems (Saputro et al., 2015; Welgama and Gibson 1995). The expert system concept and decision support approach have benefited from technological advancements in software packages. A number of software solutions advising on MHES were available up to 2005 (Saputro et al., 2015).

Optimization approach aims to find the best possible solution to a selection problem for any given mathematical model, whether it is single-objective or multi-objective, within any given set of constrains (Saputro et al., 2015). With well-quantified and well-measured information and system conditions, optimization approaches become feasible (ibid). The hybrid approach combines two distinct approaches, for example, AI and optimization (Mirhosseyni et al., 2009; Welgama and

Gibson 1995), AI and MCDM (Chan et al., 2001; Park, 1996), MCDM and optimization (Braglia et al., 2001; Yilmaz and Dağdeviren, 2011) or another approach altogether. A hybrid approach aims to combine both criteria and objectives for a thorough decision-making process by seeing the problem at multi-context, compensating for the shortcomings of individual methods and allowing for more informed decision-making (Saputro et al., 2015).

Each of the mentioned approaches has advantages and disadvantages; whether one is most appropriate in a specific decision-making situation is highly context dependent (Saputro et al., 2015). A brief overview of the methods' usefulness and important aspects relates to: a. equipment type: these models have mostly concentrated on material transportation, ignoring the crucial equipment required for warehousing, identifying, communicating, controlling, and providing assistance in warehouses; b. objective: other important user demands including security, enhanced productivity, equipment utilization, environmental effect, and life cycle concerns have been neglected in favor of conventional cost reduction as the main purpose; c. Subjectivity: There has been minimal focus on endogenous criteria weight estimation in relation to the utility function that assesses equipment alternatives, and available techniques generally equate subjectivity with decision makers' preferences rather than addressing it rigorously; d. Uncertainty: frequently disregarded is the fact that certain equipment attributes are stochastic; e. Ordinal and cardinal criteria cannot be processed simultaneously by numerous models; f. More efforts has to be put into creating MHES tools that are systematic, thorough, and consistent, with a focus on how well they work in practice (Saputro et al., 2015).

Instead of using analytical models and expert systems, decision-makers depend on their own judgement, expertise, and knowledge with certain pieces of equipment. However, this method is biassed towards using certain kinds of equipment (Matson et al., 1992). Rule of thumb and basic tools like decision trees, spreadsheets, and matrices are also available to decision makers (Baker and Canessa, 2009). In the current state of MHES approaches, only a quarter of them can handle both quantitative and qualitative data at the same time, while the other three quarters rely only on quantitative data (Khandekar and Chakraborty, 2015). Recently, approaches to improving the consistency of findings and the management of uncertainty and preference information have been developed via the integration of stochastic multicriteria acceptability analysis (SMAA) with other methodologies (Garcia et al., 2022).

According to several studies (Tuzkaya et al., 2010; Mirhosseyni and Webb, 2009; Chan et al., 2001; Chu et al., 1995; Matson et al., 1992; Prasad et al., 2015), an effective MHES can increase financial return, decrease response times, rationalize the use of human resources, and make the productive system more flexible. Making choices to reduce move lengths, increase handled unit size, search out possibilities for circular trips during product storage or order selection, and improve cube utilization are all part of selecting the proper equipment for the handling of materials (Manzini, 2012).

The MHES decision criteria that have been identified are shown in Figure 1. A complicated decision problem arises when trying to determine the optimal mix of these and other aspects of the process (Onut et al., 2009). However, the MHES process is able to identify the specific equipment needs by carefully managing this complexity.

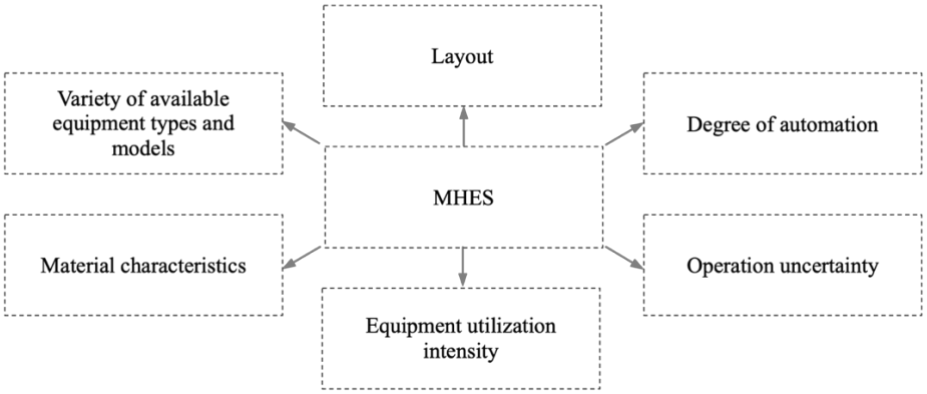


Figure 5. MHES decision criteria. Source: Carcia et al., 2022

3.4 Human-centered approaches in industrial design and decision-making processes

Human-centered approaches in industrial design and decision-making processes stress the need of including stakeholders and employees in the design and implementation phases. Incorporating qualitative insights generated from the actual experiences of those who interact with the systems daily challenge the traditional dependence on quantitative data alone. The inclusion of human factors ensures that the systems and processes developed are not only efficient but also user-friendly and adaptable to real-world challenges (Kotter, 1996; Maguire, 2001).

Change procedures can only be effectively implemented when employees are actively involved. This fosters a sense of ownership and commitment, which in turn leads to better and more long-lasting results (Kotter, 1996). This viewpoint is in line with the ideas of human-centered design, which stress the need of including end-users from the very beginning. Solutions can be better adapted to users' real needs when they are included in the early phases of system design; this increases their adoption and efficacy (Maguire, 2001).

Various methods support human-centered design, emphasizing that systems designed with user input are more likely to be effective and well-received (Maguire, 2001). Introducing human aspects into system design is essential for reducing human error and increasing safety and efficiency in industrial settings, where jobs can be complicated and human mistake is a real possibility (Reason, 2000). This method takes into account the technical parts of system design as well as the real, everyday problems encountered by people working with these systems (Reason, 2000).

To keep the systems up-to-date and efficient throughout time, employee participation is crucial, and this is true even after the design phase has begun (Bergman et al., 2015). When employees actively participate in process improvement projects, their insights from the operations are taken into account, leading to more durable results. This, in turn, promotes a culture of continuous improvement (ibid).

More effective and relevant results are produced through the design process when both staff and end-users are actively participating in the co-creation concept (Sanders & Stappers, 2008). By involving the users at every stage, this collaborative method guarantees that the end product will satisfy both operational needs and user expectations (ibid).

The design process yields sustainable and user-centred outcomes when stakeholder insights, including those of employees, are integrated. This method of strategic design places an emphasis on people while making systems, so they may be both useful and flexible enough to meet the organization's evolving demands (Manzini & Vezzoli, 2003). The importance of human-centered approaches in industrial design is demonstrated by the fact that worker participation in process redesign can result in substantial productivity increases (Adler, 1993).

4 CASE COMPANY

This chapter provides a comprehensive overview of HAMC's current operations, aiming to define the focus area. It offers a look into HAMC's logistics operations, processes, and the material handling equipment (MHE) currently utilized. Empirical insights were gathered through a thorough review of documents available on HAMC's SharePoint platform, as well as through interviews and observations with employees and managers across various departments, including outbound operations, packaging, and buffering areas. This detailed exploration lays the groundwork for analysing the existing material handling system and identifying areas for improvement.

4.1 Introduction to HAMC's product portfolio

HAMC offers a wide variety of home appliances including major household appliances (MDA), small domestic appliances (SDA), Hobs, and accessories for MDAs. MDA category includes a range of product for cooking and refrigeration such as freezers, refrigerators, stoves, ovens, and microwaves, as well as products for washing and drying such as laundry machines, dryers, and dishwashers.

SDA category includes a range of products for cooking and cleaning such as electric kettles, coffeemakers, hoover cleaners, hot plates and other equipment used for cooking and cleaning. Another important part of HAMC's product portfolio is their range of hobs.

Along with appliances, HAMC also provides a variety of accessories that are specifically made to work with MDA goods, these add-ons include things like racks for ovens and refrigerators, dishwasher baskets, and hoses for washing machines.

4.2 Overview of HAMC's process flows that demand the use of MHE

In an effort to maximize productivity, HAMC has developed and is constantly improving internal process models. Utilization of MHE is fundamental to the incoming, outgoing, and replenishing procedures. In order to identify the different MHE needs, it was crucial to gain an overview understanding of these process models. The various processes related to MHE use were mapped out by accessing the operations department's SharePoint platform. Multiple levels of information were defined for each procedure. After the process models were analyzed and studied, information about the role of MHE in each process was gathered. This knowledge allowed for a more thorough investigation into the details, which in turn allowed for the identification of problem areas.

Incoming tasks that demand for MHE utilization

Depending on the product category being handled, HAMC's incoming tasks that demand the use of MHE might vary. For MDA, tasks include unloading products clamp by clamp, sorting them (including physically sorting out damaged products and reporting damages), staging them by vendor item barcode (VIB), scanning them by VIB (and reporting any over/under delivery), taking them from the source bin to the destination bin, and finally, confirming the destination bin. Figure 6 presents part of the MDA incoming process flow that demand for MHE utilization.

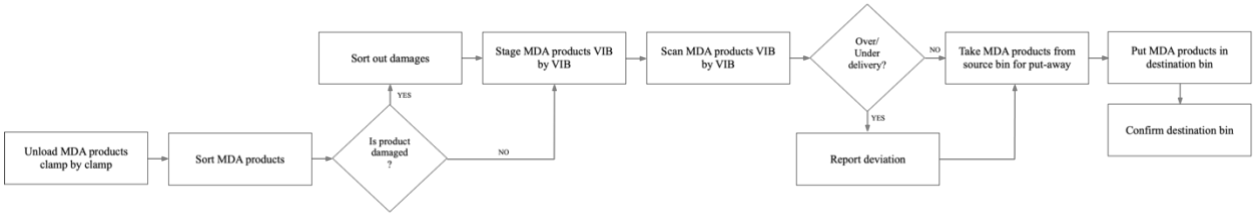


Figure 6. Part of HAMC's MDA incoming process flow that demand for MHE utilization.

Tasks for SDA include unloading and staging the products pallet by pallet, scanning the products, printing and applying pallet labels, taking the products from the source bin to the destination bin, and confirming the destination bin. HAMC also have procedures for separating products on pallets and reporting damaged products. Figure 7 presents part of the SDA incoming process flow that demand for MHE utilization.

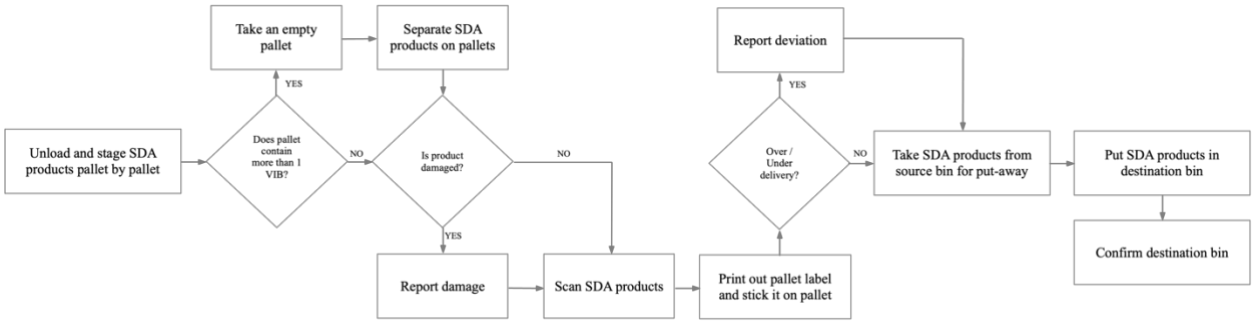


Figure 7. Part of HAMC's SDA incoming process flow that demand for MHE utilization.

Outgoing tasks that demand for MHE utilization

Outgoing HAMC activities requiring MHE could differ, similar to incoming tasks, based on the product type. For MDA appliances, a clamp truck is used to pick appliances from source bins, stage them and check for completeness. Any deviations are reported to the coordinator so that they may be addressed. After the verification is complete, the coordinator oversees the printing of loading paperwork and the loading of appliances into outgoing trucks. Part of HAMC's MDA outgoing process flow that demand for MHE utilization is shown in figure 8.

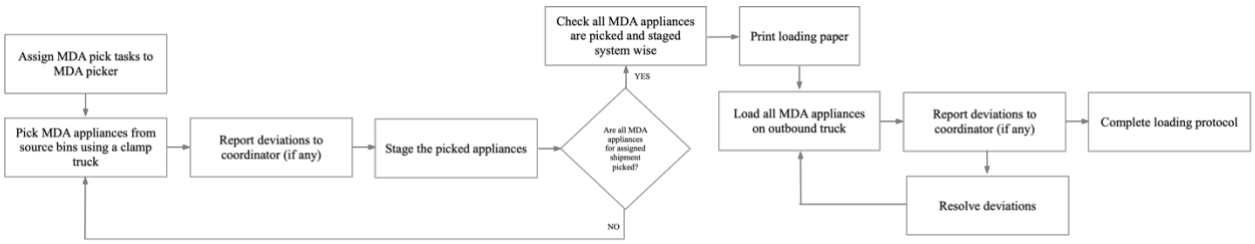


Figure 8. Part of HAMC's MDA outgoing process flow that demand for MHE utilization

For SDA appliances similar activities are carried out, with the assigned items being picked from source bins on pallets. Before being staged for shipment, any deviations are reported, and pallets are checked for completeness. Appliances are either immediately staged for transport or packaged into boxes and palletized according to packaging requirements. The coordinator handles any deviations after staging, loading papers are printed, and pallets are loaded onto outbound trucks. Figure 9 presents part of HAMC's SDA outgoing process flow that demand for MHE utilization.

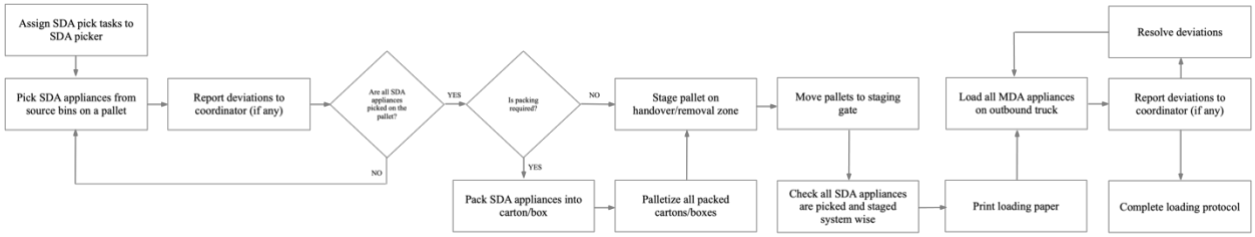


Figure 9. Part of HAMC's SDA outgoing process flow that demand for MHE utilization.

4.3 Overview of HAMC's existing MHE

A fleet of specialized industrial trucks support HAMC's warehouse operations. The trucks are strategically deployed over the three levels of the warehouse: ground, first (M1), and second (M2), in order to carry out important tasks such as unloading, loading, picking, put-away, and replenishing.

Electric counterbalance forklifts handle indoor and outdoor loading and unloading tasks. High reach trucks are tasked with the incoming put-away, outgoing staging, and replenishment of MDA, SDA, and HOBS. These high reach trucks can lift items up to 10 meters in height.

Spread out over many levels, low-lift order picking trucks handle outgoing SDA picking and pallet trucks manage the diverse activities performed for SDA and HOBS on different levels. On M1, pallet stacking trucks handle empty pallets, and finally pallet loading and unloading batteries for charging in the charging area.

These industrial trucks are notable for their specific configuration, which includes normal and double forks for loading and unloading full SDA pallets, specialized carton clamps for handling non-palletized SDA and HOBS products, specialized carton clamps for loading, unloading, storing, and picking MDA products, and a specialized rotating arms clamp for unloading horizontally placed MDA products. Figure 10 presents the different clamps manufactured especially for HAMC.

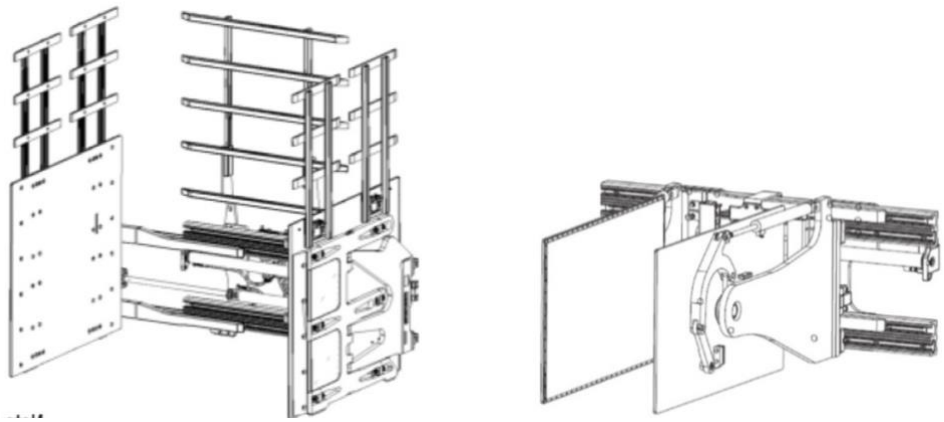


Figure 10. On the lest, 3-wide carton clamp. On the right, rotating arms carton clamp. Source: HAMC

5 EMPIRICAL FINDINGS

This chapter presents the empirical findings derived from the data collected through interviews, observations, and document reviews at HAMC. The analysis focuses on identifying the key issues and inefficiencies within the current material handling system (MHS). The purpose of this section is to provide a clear understanding of the operational challenges and opportunities for improvement by systematically categorizing, cleaning, and consolidating the collected data.

The structure of this section is organized around the types of data collected and the issues identified, rather than the specific analytical methods used. This approach allows for a comprehensive presentation of the current state of MHE utilization and the problems encountered within the system. The findings are then used as a critical input for the improvement work that follows, helping to shape the design of a new and improved MHS.

The empirical data presented here is not merely descriptive; it serves as a foundational element in the subsequent improvement work. These findings played a crucial role in guiding the design process by offering insights into demand patterns, operational inefficiencies, and employee feedback. Through a comprehensive understanding of the real-world data, combined with employee input, it was possible to make informed decisions regarding necessary changes and to evaluate which employee suggestions were feasible and cost-effective to implement.

5.1 Overview of HAMC's existing MHE utilization records

HAMC relies on a third-party software to collect utilization records of its fleet; the extracted records focus on the period 2017 to 2023. These records provided critical insights into the fleet's performance and usage patterns. The utilization records contained various metrics, such as percentage of activity usage (driving, lifting, forwards, and backwards movement), distance driven (total, forwards, and backwards), availability (idle, maintenance, and usage time), and consumption.

All data were categorized by truck type and number, allowing for a detailed analysis and comparison. Until this project, these utilization records had not been accessed or analyzed at HAMC. Therefore, an excel tool was developed to clean and compile these records for further analysis and visualization. The compiled data parameters included comprehensive information as listed below on Table 4.

Data Parameters	Data Parameters
- Truck nr	- Average Distance Driven (km)
- Model	- Average Distance Driven Outbound (km)
- Total Rental Period Cost (SEK)	- Average Distance Driven Outbound (km)
- Cost (SEK/hour)	- Consumption (kWh/km)
- Operating Hour (hours)	- Average Consumption (kWh/km)
- Average Operating Hours (hours)	- Average Consumption Outbound (kWh/km)
- Average Operating Hours Outbound (hours)	- Average Consumption Outbound (kWh/km)
- Average Operating Hours Outbound (hours)	- Department (Outbound, Outbound)
- Idle Hour (hours)	- Configuration (MDA, SDA)
- Average Idle Hours (hours)	- Floor (0, 1, 2)
- Average Idle Hours Outbound (hours)	- Drive Type (EC)
- Average Idle Hours Outbound (hours)	- Battery Technology (Lead-acid, Li-ION)
- Operating Hours Cost (SEK)	- Load Capacity (Tones)
- Average Operating Hours Cost (SEK)	- Speed (km/h)
- Idle Hours Cost (SEK)	- Access Control (RFID, PIN)
- Average Idle Hours Cost (SEK)	- Auto Logout (minutes)
- Distance Driven (km)	

Table 4. Utilization Analysis Parameters

Upon analysis and visualization of data, it became clear that significant underutilization existed across various MHE truck types, as illustrated in Figure 11. These findings provided a baseline for understanding how the current MHE was being used and where inefficiencies were occurring, which directly informed the redesign of the MHS.



Figure 11. Utilization analysis of existing fleet.

5.2 Overview of identified problems of HAMC's existing MHE

Based on interviews, observations and analysis of secondary documentation, it became clear that HAMC is dealing with numerous major problems with its current fleet of MHE. These problems were categorized into key areas: Workers' Wellbeing, Product Quality, Inefficient Utilization of Equipment, and Workflow Fluctuation. The identification of these problems was crucial in highlighting specific areas that required improvement in the new MHS design.

Workers' Wellbeing

Several reports of near-miss accidents have been associated with the lack of safety features in the fleet. As an example, an MHE operator recounted a near-miss incident that occurred during a regular unloading operation in the waggon area involving a machine that lacked protection bars and warning lights. When they unload the waggons, they keep all the cargo on the dock until they finish counting down and sorting, and then they start putting everything away. However, he nearly collided with another forklift because there wasn't enough visibility or warning flags in that location. A picker also brought attention to the fact that the forklifts did not have safety belts in another incident report. She described a situation where the operator nearly fell off the forklift while handling a big cargo; the operator should have worn safety belts to avoid the accident.

On top of that, workers' concentration tends to decrease and accident risk rises when they get used to doing the same things over and over again. This was brought to light in a specific event that the logistics manager reported. He saw that staff frequently disregarded safety rules because they were overconfident. He related an incident where an operator disregarded regular operating protocols at a busy period, which resulted in an accident where the operator sprang out of a moving MHE and the machine crushed his foot.

In addition, the fluctuations of high and low traffic periods rendered many personnel inactive for extended periods. Because their skills and memory had deteriorated from inactivity, the likelihood of mishaps increased when operations began. Operators returning from extended periods of inactivity had trouble remembering safety measures, leading to an increase in near-miss accidents, according to a team leader. A major incident occurred when an operator almost caused the stack to collapse because they forgot the proper way to retrieve MDA from high bins.

Product Quality

Significant issues, including allegations of smuggling, have resulted from certain HAMC vehicles lacking essential equipment, such as barcode scanners or order processing consoles. As an example, the quality operations manager found that some loading and unloading vehicles were missing the necessary equipment, which led to the improper processing of many shipments during an audit. The mistake caused goods to be sent to the incorrect nation, which in turn led to customs officials accusing the shipper of smuggling. Accurate shipping tracking and verification is difficult, he said, without the right scanning equipment.

Damage to products from careless use of equipment or failure to adhere to standard operating procedures is another factor that lowers product quality. According to the quality operations manager, there was an incident where a forklift operator, who had not received adequate training, severely damaged a batch of MDA. He detailed how the operator cranked the forklift too hard while trying to put MDA on top of another stack, crushing and rendering the products useless.

In another instance, the quality operations manager detailed an incident that occurred during the observation study. It involved warehouse workers who, pressed for time, disregarded standard operating procedures, leading to the incorrect storage of items—which could have resulted in drug charges.

These concerns highlight the importance of having the right tools and following the rules to the letter in order to keep product quality high and avoid additional logistical issues.

Inefficient Utilization of Equipment

High maintenance costs, rapid energy loss, and lengthy downtimes are the outcomes of this. For instance, a senior forklift operator documented multiple incidents when employees utilised material handling equipment (MHE) meant for large loads to transfer empty pallets. The power of the forklifts was diminished by this procedure. In other instances, he found that operators would use MHE meant for lighter weights to move heavier pallets. This would result in the MHE going out of commission for days, which would increase maintenance expenses and cause operations to be disrupted.

The improper interaction of the equipment with other warehouse equipment, like ramps, lifts, and wrappers, also frequently causes damage to the equipment. The manager of safety and maintenance brought attention to multiple instances where loading ramps were damaged by forklifts due to the improper handling of certain goods in terms of weight and size. He gave the example of an operator who caused expensive repairs after crashing into the gate while trying to load MDA with an unloading MHE.

Another time, a team leader saw that some employees were abusing MHE, which was causing problems with their workflow. He described an incident where packing personnel attempted to use picking MHE as a lifting machine to speed up the wrapping process, but this led to a breakdown in the picking workflow. He stressed that the inefficient use of equipment raised lease costs, decreased equipment longevity, and drastically diminished warehouse operations' overall efficiency.

The requirement for correct training and compliance with equipment usage protocols is demonstrated by these instances, which show how operating costs and efficiency can be negatively impacted by equipment repurposing and misuse

Workflow Fluctuation

Due to inflexibility and the wrong distribution of truck types within the warehouse, the current fleet arrangement creates significant delays. Warehouse employees' efforts to put vehicles to use for tasks they weren't meant for or repurpose equipment only make the problems worse. During peak moments, operators frequently utilise high-reach trucks specifically designed for stacking pallets to transfer products across the warehouse floor, according to a team leader. The high-reach trucks are slowed down and the aisles get congested due to this usage, which creates major bottlenecks in the operation.

Even worse, when a lot of warehouse workers aren't doing anything, the process becomes inefficient and delayed. This can be because other coworkers are already doing the work or because they lack the necessary tools for the job. Several employees were hindered from carrying out their tasks because there weren't enough loading trucks, according to a shift supervisor. Order fulfilment was significantly delayed because personnel were left idle while the necessary equipment became available.

In addition, it is not uncommon for equipment to malfunction when used for an unforeseen purpose. For example, the transport operations manager had vast expertise as an operator for many years. He once witnessed an event where forklifts intended for loading and unloading were used for putaway and staging, leading to a malfunction and costly repairs. Not only did this slow down production, but it also raised maintenance expenses and cut into equipment availability for other jobs.

These cases show how inefficient workflows and changes in truck types are caused by equipment reuse and inappropriate distribution. If we want to make warehouse operations more efficient and

productive overall, we need to fix these problems by making sure the right equipment is available and used correctly.

5.3 A High-Level Review of the Volumes Entering and Exiting HAMC

Understanding the volume of products entering and exiting HAMC's warehouse was essential for designing an efficient MHS. The analysis of inbound and outbound volumes provided insights into demand patterns, peak periods, and resource allocation needs. The data collected here served as the foundation for determining the capacity requirements and operational strategies for the new MHS.

Among the products handled by HAMC are MDA, SDA, HOBS, and accessories. The detailed examination of volume distributions across different product categories and time periods helped identify specific operational bottlenecks and informed decisions about the necessary types and quantities of MHE. This data-driven approach ensured that the new MHS design would be robust enough to handle current and future demands efficiently. The analysis was divided into two sections, inbound and outbound volumes.

Inbound Volume Analysis

The inbound volumes analysis focused on the period from 04/ 2022 to 12/2023, with the extracted data reaching up to 90.000 records. An excel tool was developed to clean and compile the data, resulting in a final dataset of 7.000 records. Table 5 and 6 illustrates the parameters analyzed and the outcomes of the analysis, which were later used as inputs for the new design of the MHES.

Data Parameters	Data Parameters
- Date	- Total SDA Pieces
- Day	- Total SDA Volume (m^3)
- Means of Transport ID	- Total HOBS Pieces
- Means of Transport (Container, Truck, Wagon)	- Total HOBS Volume (m^3)
- Country of Origin	- Equipment Needs (Counterbalance Truck, Electric Pallet Truck, High Reach Truck)
- Material Category (MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, MDA/SDA/HOBS)	- Booked Unloading Slot Time
- Total Pieces	- Unloading Start Time
- Total Volume (m^3)	- Unloading Finish Time
- Total MDA Pieces	- Put-away Start Time
- Total MDA Volume (m^3)	- Put-away Finish Time

Table 5. Inbound Volume Analysis Parameters

Analysis	Outcomes
Daily volume handled by containers, trucks, wagons, and total inbound	Average (m^3/day), Standard deviation (m^3/day), Max (m^3/day), Min (m^3/day), 75% Quartile (m^3/day)
Daily number of containers, trucks, wagons, and total inbound handled by HAMC	Average (m^3/day), Standard deviation (m^3/day), Max (m^3/day), Min (m^3/day), 75% Quartile (m^3/day)
Daily distribution of total number of containers, trucks, wagons, and total means of transport handled by HAMC	Total number of containers, trucks, wagons, and total inbound means per weekday
Daily percentage distribution of total number of containers, trucks, wagons, and total inbound handled by HAMC	Percentage of containers, trucks, wagons, and total inbound per weekday
Daily distribution of distinct number of containers, trucks, wagons, and total inbound handled by HAMC (How many times a specific weekday accommodated the specific type of means of transport)	Distinct number of containers, trucks, wagons, and total inbound per weekday
Daily distribution of average number of containers, trucks, wagons, and total inbound handled by HAMC	Average number of containers, trucks, wagons, and total inbound per weekday
Daily distribution of maximum number of containers, trucks, wagons, and total inbound handled by HAMC	Maximum Number of containers, Trucks, Wagons, and Total Inbound per Weekday
Daily distribution of minimum number of containers, trucks, wagons, and total inbound handled by HAMC	Minimum number of containers, trucks, wagons, and total inbound per weekday
Daily distribution of total volume delivered by containers, trucks, wagons, and total inbound at HAMC	Total volume delivered by containers, trucks, wagons, and total inbound per weekday
Daily percentage distribution of total volume delivered by containers, trucks, wagons, and total inbound at HAMC	Percentage of total volume delivered by containers, trucks, wagons, and total inbound per weekday
Daily distribution of average volume delivered by containers, trucks, wagons, and total inbound at HAMC	Average volume delivered by containers, trucks, wagons, and total inbound per weekday
Daily distribution of maximum volume delivered by containers, trucks, wagons, and total inbound at HAMC	Maximum volume delivered by containers, trucks, wagons, and total inbound per weekday

Analysis	Outcomes
Daily distribution of minimum volume delivered by containers, trucks, wagons, and total inbound at HAMC	Minimum volume delivered by containers, trucks, wagons, and total inbound per weekday
Monthly distribution of total volume delivered by containers, trucks, wagons, and total inbound at HAMC	Total volume delivered by containers, trucks, wagons, and total inbound per month
Monthly percentage distribution of total volume delivered by containers, trucks, wagons, and total inbound at HAMC	Percentage of total volume delivered by containers, trucks, wagons, and total inbound per month
Monthly distribution of distinct number of containers, trucks, wagons, and total inbound handled by HAMC (How many days in a month HAMC accommodated the specific type of means of transport)	Distinct number of days HAMC accommodated , trucks, wagons, and total inbound per month
Monthly distribution of total volume per product category	Total volume of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS, MDA/SDA/HOBS, and total inbound per month
Monthly distribution of total number per product category (How many times in a month HAMC received the specific product category)	Total number of times HAMC received MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, MDA/SDA/HOBS, and total inbound per month
Daily distribution of total volume per product category delivered at HAMC	Total volume of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS,MDA/SDA/HOBS, and total inbound per weekday
Daily distribution of total number per product category delivered at HAMC (How many times in a specific weekday HAMC received the specific product category)	Total number of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS,MDA/SDA/HOBS, and total inbound per weekday
Distribution of total volume per product category per means of transport delivered at HAMC (How much volume of a specific product category HAMC received by a specific means of transport in the entire period)	Total volume of MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, and MDA/SDA/HOBS per container, truck, and wagons

Analysis	Outcomes
Distribution of total number of product category per means of transport delivered at HAMC (How many times a mean of transport delivered products of a specific product category at HAMC in the entire period)	Total number of MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, and MDA/SDA/HOBS per container, truck, and wagons
Hourly distribution of total volume delivered by containers, trucks, wagons, and total inbound at HAMC	Total volume delivered by containers, trucks, wagons, and total inbound per hour
Hourly distribution of average volume delivered by containers, trucks, wagons, and total inbound at HAMC	Average volume delivered by containers, trucks, wagons, and total inbound per hour
Hourly distribution of standard deviation of average volume delivered by containers, trucks, wagons, and total inbound at HAMC	Standard deviation of volume delivered by containers, trucks, wagons, and total inbound per hour
Hourly distribution of maximum volume delivered by containers, trucks, wagons, and total inbound at HAMC	Maximum volume delivered by containers, trucks, wagons, and total inbound per hour
Hourly distribution of minimum volume delivered by containers, trucks, wagons, and total inbound at HAMC	Minimum volume delivered by containers, trucks, wagons, and total inbound per hour
Hourly distribution of total number of containers, trucks, wagons, and total inbound handled by HAMC	Total number of containers, trucks, wagons, and total inbound that HAMC handled per hour
Hourly distribution of average number of containers, trucks, wagons, and total inbound handled by HAMC	Average number of containers, trucks, wagons, and total inbound that HAMC handled per hour

Table 6. Inbound Volume Analysis Outcomes

Gathering detailed information on the hourly distribution of volume, means of transport, and product categories was a challenge, due to the different tools used by HAMC for scheduling processes and the absence of a central database to integrate all relevant data into a single file.

Hourly distributions could not be accessed in real-time. As an alternative approach, an earlier study on inbound workload balancing was cited, since it had some relevant hourly distribution data statistics. While this information may not been fully current, it served as a valuable reference point to offer insights into the hourly distribution patterns and guide initial assessments.

The analysis of inbound operations provided a comprehensive understanding of distribution densities across various dimensions such as the average daily inbound volume and the standard deviation, transportation dynamics based on the average number of means of transport per day,

temporal patterns, including daily and monthly distribution densities, highlighted peak activity on specific days and months. Notably, after the analysis emerged peak days for means of transport, total volume, and average volume.

Furthermore, the product category distribution for total volume emphasized the dominance of categories such as which product category demands for more resources than others. The analysis revealed a significant difference in hourly distribution densities for the average number of means, emphasizing heightened operational demands on specific timeslots during the day. Moreover, the analysis revealed the number of the necessary trucks needs to successfully complete all operations. This comprehensive examination provided detailed understanding of HAMC's inbound dynamics, and a robust foundation for strategic decision-making during the design phase.

Outbound Volume Analysis

The outbound volumes analysis focused on the period from 11/ 2022 to 11/2023, with the extracted data reaching up to 8.000 records. An excel tool was developed to clean and compile the data, resulting in a final dataset of 6.800 records. Table 7 and 8 illustrates the parameters analyzed and the outcomes of the analysis, which were later used as inputs for the new design of the MHES.

Data Parameters	Data Parameters
- Date	- Counter Balance Truck (Yes/No)
- Day	- Electric Pallet Truck (Yes/No)
- Means of Transport (Container, Truck, Wagon)	- Horizontal Picker Truck (Yes/No)
- Carrier	- High Reach Truck (Yes/No)
- Shipment Number	- Booked Loading Timeslot
- Country	- Actual Arrival Time
- Material Category (MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, MDA/SDA/HOBS)	- Loading Start Time
- Total Pieces	- Loading Finish Time
- Total Volume (m^3)	- Total Loading Time
- Total MDA Pieces	- Picking Start Time
- Total SDA Pieces	- Picking Finish Time

Table 7. Outbound Volume Analysis Parameters

Part of the outcomes of this excel tool was a visual map of the loading dynamics pattern as shown in Figure 12.

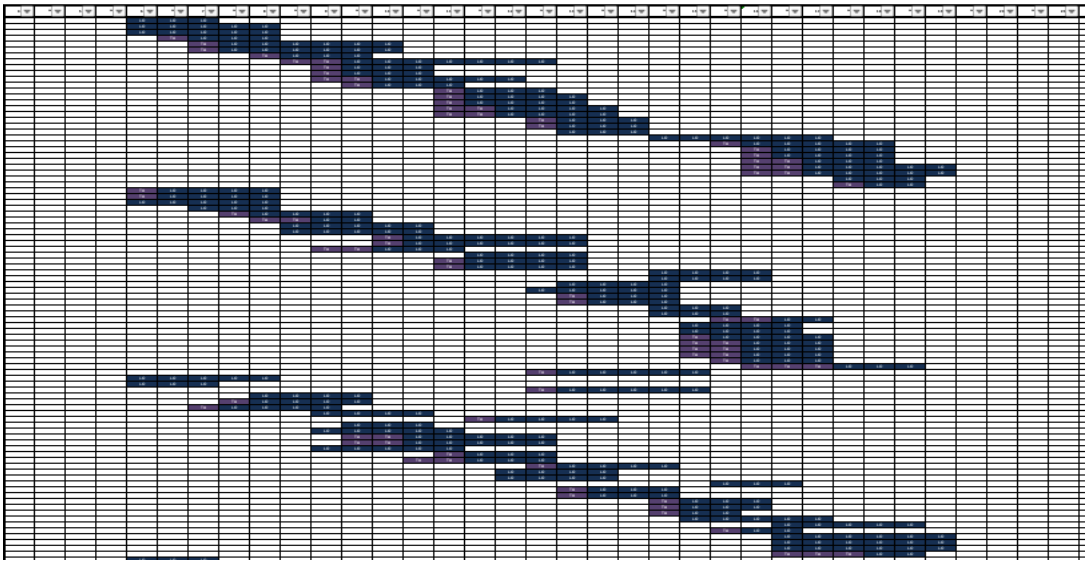


Figure 12. Visual map of the loading dynamics pattern

Analysis	Outcomes
Daily volume handled by trucks, wagons, and total outbound	Average (m^3/day), Standard deviation (m^3/day), Max (m^3/day), Min (m^3/day), 75% Quartile (m^3/day)
Daily number of trucks, wagons, and total outbound handled by HAMC	Average (m^3/day), Standard deviation (m^3/day), Max (m^3/day), Min (m^3/day), 75% Quartile (m^3/day)
Daily distribution of total number of trucks, wagons, and total means of transport handled by HAMC	Total number of trucks, wagons, and total outbound means per weekday
Daily percentage distribution of total number of trucks, wagons, and total outbound handled by HAMC	Percentage of trucks, wagons, and total outbound per weekday
Daily distribution of distinct number of trucks, wagons, and total outbound handled by HAMC (How many times a specified mean departed from HAMC on a given day)	Distinct number of trucks, wagons, and total outbound per weekday
Daily distribution of average number of trucks, wagons, and total outbound handled by HAMC	Average number of trucks, wagons, and total outbound per weekday
Daily distribution of maximum number of trucks, wagons, and total outbound handled by HAMC	Maximum Number of Trucks, Wagons, and Total Outbound per Weekday

Analysis	Outcomes
Daily distribution of minimum number of trucks, wagons, and total outbound handled by HAMC	Minimum number of trucks, wagons, and total outbound per weekday
Daily distribution of total volume departed by trucks, wagons, and total outbound from HAMC	Total volume departed by trucks, wagons, and total outbound per weekday
Daily percentage distribution of total volume departed by trucks, wagons, and total outbound from HAMC	Percentage of total volume departed by trucks, wagons, and total outbound per weekday
Daily distribution of average volume departed by trucks, wagons, and total outbound from HAMC	Average volume departed by trucks, wagons, and total outbound per weekday
Daily distribution of maximum volume departed by trucks, wagons, and total outbound from HAMC	Maximum volume departed by trucks, wagons, and total outbound per weekday
Daily distribution of minimum volume departed by trucks, wagons, and total outbound from HAMC	Minimum volume departed by trucks, wagons, and total outbound per weekday
Monthly distribution of total volume departed by trucks, wagons, and total outbound from HAMC	Total volume departed by trucks, wagons, and total outbound per month
Monthly percentage distribution of total volume departed by trucks, wagons, and total outbound from HAMC	Percentage of total volume departed by trucks, wagons, and total outbound per month
Monthly distribution of total number of trucks, wagons, and total outbound departed from HAMC	Distinct number of days per month trucks, wagons, and total outbound departed from HAMC
Monthly distribution of distinct number of trucks, wagons, and total outbound departed from HAMC (How many days in a month a specific type of means of transport departs from HAMC)	Distinct number of days per month trucks, wagons, and total outbound departs from HAMC
Monthly distribution of total volume per product category	Total volume of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS, MDA/SDA/HOBS, and total outbound per month
Monthly distribution of total number per product category (How many times in a month HAMC dispatched the specific product category)	Total number of times HAMC dispatched MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, MDA/SDA/HOBS, and total outbound per month

Analysis	Outcomes
Daily distribution of total volume per product category dispatched by HAMC	Total volume of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS,MDA/SDA/HOBS, and total outbound per weekday
Daily distribution of total number per product category dispatched by HAMC (How many times in a specific weekday HAMC dispatched the specific product category)	Total number of MDA, SDA, HOBS, MDA/SDA,MDA/HOBS,MDA/SDA/HOBS, and total outbound per weekday
Distribution of total volume per product category per means of transport dispatched by HAMC (How much volume of a specific product category HAMC dispatched by a specific means of transport in the entire period)	Total volume of MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, and MDA/SDA/HOBS per trucks and wagons
Distribution of total number of product category per means of transport dispatched by HAMC (How many times HAMC dispatched the specific product category with the specific mean of transportation in the entire period)	Total number of MDA, SDA, HOBS, MDA/SDA, MDA/HOBS, and MDA/SDA/HOBS per truck, and wagons
Hourly distribution of total volume HAMC dispatched by trucks, wagons, and total outbound	Total volume dispatched by trucks, wagons, and total outbound per hour
Hourly distribution of average volume HAMC dispatched by trucks, wagons, and total outbound	Average volume dispatched by trucks, wagons, and total outbound per hour
Hourly distribution of standard deviation of average volume HAMC dispatched by trucks, wagons, and total outbound	Standard deviation of volume dispatched by trucks, wagons, and total outbound per hour
Hourly distribution of maximum volume HAMC dispatched by trucks, wagons, and total outbound	Maximum volume dispatched by trucks, wagons, and total outbound per hour
Hourly distribution of minimum volume HAMC dispatched by trucks, wagons, and total outbound	Minimum volume dispatched by trucks, wagons, and total outbound per hour
Hourly distribution of total number of trucks, wagons, and total outbound dispatched by HAMC	Total number of trucks, wagons, and total outbound that HAMC dispatched per hour

Analysis	Outcomes
Hourly distribution of average number of trucks, wagons, and total outbound dispatched by HAMC	Average number of trucks, wagons, and total outbound that HAMC dispatched per hour

Table 8. Outbound Volume Analysis Outcomes

The analysis of outbound operations provided a comprehensive understanding of distribution densities across various dimensions such as the average daily outbound volume and the standard deviation, transportation dynamics based on the average number of means of transport per day, temporal patterns, including daily and monthly distribution densities, highlighted peak activity on specific days and months. Notably, after the analysis emerged peak days for means of transport, total volume, and average volume.

Furthermore, the product category distribution for total volume emphasized the dominance of categories such as which product category demands for more resources than others. The analysis revealed a significant difference in hourly distribution densities for the average number of means, emphasizing heightened operational demands on specific timeslots during the day. Moreover, the analysis revealed the number of the necessary trucks needs to successfully complete all operations.

6 IMPROVEMENT EXPLORATION

This chapter provides a comprehensive overview of the theoretical and practical concerns based on empirical data analysis that contributed to the design of the new MHE fleet for HAMC, as well as the improvement work that was done. The structure of this chapter is intentionally designed to create a clear linkage between the empirical findings and the improvement work, which are closely interconnected. This data was not merely descriptive but served as the foundational input for the improvement processes that followed.

In the improvement work section, the focus shifts from the raw data collection and analysis to the application of these findings in the design process. The analyzed data, which was systematically gathered, cleaned, and consolidated, was crucial for understanding demand patterns, operational inefficiencies, and employee feedback. These insights guided the improvement work, allowing for informed decisions on what changes were necessary and which employee suggestions were feasible and cost-effective to implement.

The first section of this chapter describes the approach used in the improvement work, going into depth on how the MHE fleet was improved by using the principles of Material Handling Equipment Selection (MHES). The empirical data provided the basis for understanding the existing system's shortcomings and identifying opportunities for improvement. The analysis of the utilization records, the volume fluctuation records, and the outcomes of the observation study and the interviews ensured that the design evolved in response to both data-driven insights and stakeholder input.

The second section focuses on the specific improvements made to HAMC's existing MHE fleet, illustrating how these changes were directly informed by the empirical findings. By classifying improvements according to MHE type, the section highlights how data-driven standards were applied to create a more efficient and versatile fleet. The decision to structure the chapter in this way underscores the practical application of empirical data in the engineering process, bridging the gap between analysis and implementation.

Finally, the last section describes the final MHE fleet architecture, explaining how the improvements are expected to impact operational efficiency and cost-effectiveness. This structure not only presents the data but also demonstrates its direct influence on the engineering decisions made throughout the improvement work. By aligning empirical findings with engineering outcomes, this chapter provides a clear narrative of how data was transformed into actionable improvements, ensuring that the final design is both practical and well-informed.

6.1 Improvement work approach

The improvement work was based on a knowledge-based approach with emphasis of user involvement in an iterative design process. The stakeholders involved in this process included a diverse group of employees from various departments, each playing a crucial role in the design and refinement of the new MHE fleet. These stakeholders ranged from operational managers and team leaders to warehouse operators and packers, ensuring that insights were gathered from different levels of the organization.

The combination of interviews with relevant stakeholders, observation studies within different workstations, and analysis of MHE utilization records from 2017 until 2023, incoming and outgoing product flows for years 2022 and 2023, increased the credibility of the conducted work. The stakeholders, who were primarily warehouse employees directly interacting with the MHE fleet, provided invaluable input that guided the improvement process.

The improvement work was conducted in an iterative way, where changes were made, evaluated and then changed even further in every improvement cycle. This approach allowed the design

specification to evolve step by step. In order to perform a good MHE system design, and since there was a lack in critical data, it was important to involve stakeholders - specifically those who were warehouse employees engaged in daily operations - early and repeatedly in the development process. The improvement work had a high involvement of stakeholders where the improvements were tested by the employees after each iteration in the development cycle in order to receive new input for further improvements to be made. Another reason for involving the stakeholders in the improvement work was that HAMC was not measuring critical time-related data and thus the knowledge of the employees was an important factor to validate the efficiency of the changes. The employees were therefore included in the engineering process from the start and their opinions were used throughout the entire improvement work.

The improvements that were made to the MHE fleet were conducted on two different levels; overall MHE architecture, as well as improvements to specific MHE features. First were the improvements concerning the overall consistence of the new MHE fleet carried out; this included re-organizing the types of the industrial trucks, and creating a more versatile fleet. These improvements were tested and iterated several rotations in the improvement cycle in order to find a suitable consistence of the fleet for the employees. During the re-organization of the fleet, a need for additional features and MHE types was also identified. For these new features, specifications were created and introduced to the selection process.

The second level of improvement focused on the additional MHE types developed. These improvements included the explanations of usage of the new industrial trucks. These improvements were also iterated several rotations in the improvement cycle until they considered optimal by the employees.

The first part of the improvement work focused on how the employees were utilizing the existing MHE fleet. It was important to consider those who operate the MHE in their day-to-day work when designing the new MHE fleet and therefore were interviews conducted with many warehouse employees that perform tasks such as loading, unloading, put-away, picking, staging, packing, replenishing in order to understand how they utilize the fleet and what types of trucks and features needed to be included in the new MHE fleet in order for it to support their work. This analysis of needed trucks in the fleet and features was performed as a task analysis where the tasks and the relationships between them were described in order to understand what tasks the operators would utilize the new fleet for and how those tasks would be performed. As a result of this task analysis, it became clear which trucks that were redundant in the existing fleet and thereby could be removed, as well as what features are needed by the employees however not included in the existing trucks.

In addition, an efficiency evaluation was performed on the existing MHE fleet to measure the level of efficiency achieved. The warehouse employees were asked to evaluate the efficiency of the existing MHE fleet and specify if they had encountered any problems while working with the existing fleet. This was done to find improvement areas to work with during the improvement work on the architecture of the new MHE fleet. The results of this efficiency evaluation clearly showed that the existing MHE fleet had several efficiency problems. Significant equipment underutilization resulted in wasteful use of resources and higher operational expenses. Maintenance concerns caused frequent downtime, which disrupted workflow and decreased production. Outdated trucks that couldn't handle the current operational needs reduced overall efficiency and required human involvement. Repairs and part replacements occurred often, leading to high maintenance expenditures and increasing operational expenses overall. Modern warehouse operations rely on advanced features like automated tracking and real-time monitoring to enhance productivity and accuracy. Unfortunately, the current equipment lacked these capabilities. Some vehicles posed a greater threat to warehouse safety due to missing crucial safety elements, which raised the likelihood of accidents and near-misses.

When the architecture of the new MHE fleet was to be designed, a suggestion for the new fleet was discussed with the warehouse employees in individual meetings. The suggestion for the new fleet was introduced as a brief analysis of numbers and type of trucks per task in order to identify efficiency problems with the design before dedicate to the finalization of the new MHE system. This brief analysis of the new MHE fleet communicated to the employees that the suggestion was not a final and decided one and was easier for them to get involved in the improvement of the new fleet. After having explained the new MHE fleet in the individual meetings with the warehouse employees, a few group meetings with key operators, team leaders and managers followed where they were asked to give their feedback on the new MHE fleet and then redesign the new MHE fleet themselves in order to better suit their needs and preferences. The employees then suggested their own version of how the new MHE fleet should look like. This approach of participatory design increased the chance of the new MHE fleet meeting the needs of the operators and increase operational efficiency.

The redesign activity was performed with all key stakeholders in the quality operations department, the transport operations department, and the warehouse operations department; and the opinions and improvements from those redesigns were used as input for the next iteration of designing a new architecture for the MHE fleet. The MHE fleet architecture was improved between the meetings with the different stakeholders, thereby testing the stakeholder's different ideas on the other stakeholders. When all stakeholders had received the chance to redesign the architecture of the new MHE fleet, a thorough evaluation of their suggestions was conducted in two stages. The first evaluation was conducted as a rapid simulation evaluation where warehouse employees were asked to adjust the current MHS to simulate the new design, which allowed to observe and investigate the efficiency of the proposed solution. The second evaluation was conducted in comparison with existing MHE fleet utilization records, incoming and outgoing goods flow expectations, the literature review, the results of a market research, and the results of an equation developed to identify the minimum amount of required MHE trucks in order to improve the new MHE fleet even further.

The stakeholders were encouraged to interact extensively with each other during these meetings in order to have an open discussion about the new MHE fleet and create a shared ownership of the newly designed MHE fleet. The open discussion ensured that everyone felt that their opinions had been taken into account and that they all were satisfied with the end result. This was important since all opinions from the individual redesigns were not implemented into the new fleet as some of the stakeholders had colliding ideas. These colliding opinions were brought up and discussed during the meetings, and at the end all stakeholders could agree on a mutual MHE fleet.

6.2 Development of data analysis models

An integral aspect of the project was the development of a data analysis model to ascertain the bare minimum of necessary MHE vehicles. The current utilization records and the expected incoming and outbound product flows were used to construct this model. Overall operational efficiency was to be improved by measuring all parts of the material handling process, finding inefficiencies, and suggesting improvements.

A multi-stage process was used to build the data analysis model. Data from the quantitative study, which included records of MHE use from 2017 to 2023 and data on product flows for 2022 and 2023, formed the basis of the initial data set. We learnt a lot about the operating dynamics and limitations of the present MHE fleet from this extensive dataset.

Extra parameters were found and added to the model to guarantee a thorough and precise analysis. For a comprehensive evaluation that included all important parts of MHE operations, these parameters were vital. The main goal was to learn about product flow frequency and volume, operational capacity of current MHE, and area-specific requirements in the warehouse.

The formula used initially to estimate the minimum number of forklifts needed is as follows:

$$N = \frac{(D \times U)\,(G \times V)}{C}$$

Where:

- N = Minimum number of required forklifts
- D = Average daily demand for forklift tasks
- U = Utilization rate of existing fleet
- G = Growth factor expressed in incoming and outgoing daily volumes
- V = Average daily incoming and outgoing volumes
- C = Capacity of one forklift (tasks per day)

Each parameter in the formula plays a crucial role in ensuring the accuracy and reliability of the model. The average daily demand for forklift tasks (D) and the utilization rate of the existing fleet (U) provide a baseline understanding of current operations. The average daily demand for forklift tasks (D) is essential for understanding the daily workload, while the utilization rate (U) helps in assessing current efficiency and potential need adjustments. High utilization means fewer additional forklifts will be needed.

The growth factor (G) and average daily volumes (V) ensure that future increases in demand are accounted for. The calculation considers anticipated increases in workload by multiplying the growth factor with the average volume. Growth factor (G) ensures the fleet can handle future increases in volume, and average daily volumes (V) set the baseline measure of daily workload.

Finally, the capacity of one forklift (C) determines how many tasks a single forklift can handle, influencing the total number needed. By incorporating these parameters, the model provides a comprehensive estimate of the required forklifts, ensuring they are adaptable and efficient in managing varying operational demands.

Formulas for estimating the number of material handling equipment (MHE) such as forklifts in a warehouse environment often fall within the realm of operations research and industrial engineering. There are several methodologies and models developed for such purposes, but they are often context-specific and can vary widely based on the specific operational parameters and constraints of the facility in question.

Warehouse optimization models often incorporate parameters such as storage space, handling time, and throughput rates to determine equipment needs. These models can be quite complex and typically involve simulation or advanced mathematical programming.

The formula used for this study aligns with methodologies discussed in facilities planning and warehouse management literature, emphasizing the balance between demand and capacity while considering growth and efficiency (Tompkins et al., 2010; Heragu, 2008). By incorporating these parameters, the model provides a comprehensive estimate of the required forklifts, ensuring they are adaptable and efficient in managing varying operational demands.

6.3 Design of HAMC's new MHE fleet

The design of HAMC's new fleet focused on enhancing operational efficiency and cost-effectiveness by incorporating a diverse range of vehicles tailored to meet the specific needs of HAMC's warehouse operations. The new fleet includes a mix of electric counterbalance forklifts, high reach trucks, low-lift order picking trucks, and pallet trucks. The composition of the fleet is based on the amount of counterbalance trucks and their new configuration which will make them capable of accomplishing additional tasks from loading and unloading. Each type of vehicle was chosen based on its suitability for adopting into various tasks within the warehouse, such as loading, unloading, picking, and replenishing. This strategic selection ensures that the right equipment is available for each task, reducing downtime and improving operational flow. The combination of these vehicles provides a comprehensive solution to handle the dynamic requirements of warehouse operations, ensuring seamless and efficient processes.

The introduction of Lithium batteries has proven beneficial for both indoor and outdoor tasks, providing flexibility and efficiency in energy consumption and operational hours. Additional significant savings and efficiency gains have been realized with the new fleet design compared to the previous setup. Modern vehicles equipped with advanced technology require less maintenance, will result in lower costs. Enhanced safety features across the fleet have minimized the risk of accidents, thereby decreasing associated costs and fostering a safer work environment.

By ensuring vehicles are specifically chosen for their tasks, workflow bottlenecks have been reduced, leading to increased productivity. Overall, the new fleet design represents a strategic improvement in HAMC's material handling operations, balancing cost, efficiency, and safety.

7 DISCUSSION

This chapter, delves into the research findings, discussing their implications and contributions to the existing body of knowledge. This study offers a thorough analysis of material handling equipment (MHE) optimisation in warehouse operations by combining qualitative and quantitative methodologies. By highlighting the significance of comprehending both numerical data and human variables, the study delves into how this method connects theoretical models with real-world applications. Furthermore, the section examines the unique perspective this research adds to the literature, the gaps it addresses, and the justification for employing a mixed-methods approach. By doing so, we aim to highlight the critical insights gained from this study and their relevance in developing more effective and sustainable solutions for warehouse management.

7.1 Contribution to Literature

A more complete understanding of material handling equipment (MHE) optimisation in warehouse operations is provided by this research, which adds to the current body of literature by combining qualitative insights with quantitative data. In order to identify inefficiencies, traditional studies in this area, such as those by Gu et al. (2007) and Frazelle (2002), mostly use quantitative analysis, which focusses on numerical data. But this study goes beyond just looking at statistics; it uses interviews and observations to get to the bottom of the real issues warehouse workers confront. For instance, qualitative methods were used to discover problems including equipment misuse, the emotional toll of workflow interruptions on staff morale, and safety hazards arising from complacency and deviations from procedures. This methodology aligns with the human-centered approaches advocated by Reason (2000) and Maguire (2001), which stress the importance of integrating human factors into system design to avoid errors and improve safety and efficiency. By connecting theoretical models with real-world applications, this methodology provides a more complete picture of MHE optimisation and management. Better and more user-centric warehouse management is possible with this broader view, which takes into account the actual interactions between workers and machinery as well as the contextual elements impacting operational efficiency.

7.2 Human Factor vs. Numerical Analysis

The findings underscore the limitations of relying solely on numerical analysis in warehouse management. Studies by Tompkins et al. (2010) and Shah and Khanzode (2015) highlight the importance of efficiency metrics, but they often miss the underlying human behaviours that contribute to operational inefficiencies. Operational inefficiencies frequently stem from human behaviours such as careless equipment use, protocol violations, and safety failures induced by complacency—factors that are difficult to quantify but have significant implications for safety and efficiency. For instance, interview data did not capture near-miss incidents that occurred because some workers disregarded safety procedures in an effort to save time. Additionally, observations revealed an increase in mistake rates and a decline in operational memory after periods of inactivity. This research supports the argument made by Kotter (1996) and Bergman et al. (2015) that employee involvement and an understanding of human behaviour are crucial for effective system design and implementation. By directly engaging with employees, this study uncovers issues that traditional data-driven analyses might overlook, such as training gaps, productivity interruptions due to equipment misuse, and safety concerns. These insights are critical for developing a more comprehensive understanding of operational challenges and highlight the necessity of addressing the human element in warehouse management.

7.3 Importance of Examining Beyond Numbers

The development of efficient solutions that address both technological and human factors requires an examination of warehouse operations that goes beyond simple statistical analysis. Quantitative data can show that an issue exists, but it doesn't always tell you what the real reasons are, which are usually tied to people. Inadequate training or poor adherence to safety rules are some of the root reasons of inefficiencies that should be understood. Liker (2021) and Rother and Shook (2003), for instance, address this issue. This study backs up their findings by demonstrating that poorly trained personnel might occasionally be responsible for considerable maintenance expenditures, as documented in the literature (e.g., Gu et al., 2010). This research delves at the human factors that contribute to operational inefficiencies, including behaviour, company culture, and process adherence, by using qualitative insights. To create interventions that work and last, we need a thorough grasp of the problem at hand. Only then can we hope to find solutions that go to the bottom of things, rather than only treating the symptoms.

7.4 Added Perspective to Literature

This study contributes a fresh perspective to the literature by demonstrating the value of a mixed-methods approach in MHE optimization. By integrating quantitative data with qualitative insights, this research adds depth to the understanding of warehouse operations, highlighting the interplay between operational metrics and human factors. This perspective aligns with the arguments presented by Sanders and Stappers (2008) and Manzini and Vezzoli (2003), who advocate for human-centered and co-creative approaches in industrial design. By challenging the traditional reliance on quantitative data alone, this study emphasizes the importance of human-centered methodologies, which are crucial for achieving optimal outcomes in warehouse management. This holistic approach not only enhances the effectiveness of the solutions developed but also ensures greater buy-in from stakeholders, facilitating smoother implementation of new systems. Moreover, it underscores the risk of omitting critical contextual information when relying solely on quantitative methods, which can significantly impact the efficacy of operational improvements.

7.5 Gaps in Current Literature

The current literature on MHE optimization, as reviewed by Hassan (2010) and Saputro et al. (2015), often neglects the critical role of employee involvement in the design and implementation of material handling systems. Most studies focus primarily on numerical efficiency metrics, overlooking the practical insights that can be gained from those directly involved in daily operations. This research addresses this gap by demonstrating that user interaction is essential for identifying inefficiencies and developing workable solutions. The involvement of warehouse employees in the design process, for example, uncovered issues related to equipment misuse and workflow disruptions that would not have been evident through quantitative analysis alone. This finding aligns with the recommendations of Bergman et al. (2015), who emphasize the need for future research to conduct longitudinal studies that monitor the adoption of user-designed material handling systems and assess their long-term impact on operational efficiency. By highlighting the importance of staff participation, this study underscores the necessity of integrating practical, user-driven insights into the MHE optimization process to create more effective and sustainable systems.

7.6 Justification for Mixed Methods

Given the complex nature of warehouse operations, this study justifies the use of mixed methods, combining qualitative and quantitative assessments. This approach is consistent with the

recommendations of Creswell and Plano Clark (2018) and Tashakkori and Teddlie (2010), who argue that mixed methods provide a more comprehensive understanding of research problems by integrating numerical precision with human experience. In this study, qualitative insights revealed the contextual and behavioral factors that influence quantitative measures, offering a more nuanced understanding of operational efficiency. This approach not only enhances the robustness of the findings but also aligns with the goal of achieving more effective and sustainable solutions in warehouse management. The use of mixed methods was particularly necessary due to the lack of critical numerical data, which was supplemented by direct observations and employee feedback. This methodological choice ensures a more well-rounded analysis, combining the strengths of both qualitative and quantitative data, and ultimately leads to better-informed decisions regarding MHE optimization. By capturing the full spectrum of operational dynamics, this approach results in more effective, long-term solutions that are better suited to the complexities of warehouse environments.

8 CONCLUSIONS

This chapter provides a concise summary of the study's findings and conclusions, answering the research topic on how to optimise Material Handling Equipment (MHE) and improve the effectiveness of warehouse operations in manufacturing companies. By combining qualitative and quantitative approaches, this work significantly advances our understanding of MHE optimisation. It delves deeper than simple numerical data analysis to reveal warehouse workers' actual problems by means of interviews and observations. Workflow inefficiencies, safety risks, and the human variables impacting operational performance are just a few examples of the complex issues that these insights shed light on. A concern that was not evident in the quantitative data was that operators frequently disregarded safety rules because they believed they were short on time. This was revealed through interviews. In addition, the observations revealed that some of the equipment, although functional, was not ergonomically fit, which resulted in weariness and decreased output. The significance of utilising research methodologies that also take in consideration human factors to create better MHE solutions is shown by these findings.

Problems that are typically missed when counting numbers alone are brought to light when we consider the human element. The operational inefficiencies, such as the underutilisation of equipment during peak hours and the over-reliance on manual operations that could be automated, were revealed through employee feedback. This methodological approach provides a more comprehensive view of MHE management by connecting theoretical models with real applications. The analysis not only finds inefficiencies but also proposes practical solutions, like rearranging workstations and adding safety features, by incorporating employee feedback. Understanding the intricate nature of warehouse operations requires the use of mixed approaches, which combine qualitative insights with quantitative data. Incorporating both numerical precision and human experience, this approach delivers a balanced picture that leads to more effective and durable solutions.

By showing how beneficial a mixed-methods approach is for MHE optimisation, this study contributes a fresh viewpoint to the existing literature. In order to accomplish thorough optimisation, this study shows that it is necessary to incorporate human-centered research methodologies and challenges the customary dependence on quantitative data alone. An integrated approach can significantly increase productivity and employee happiness. The importance of employee input and participation in MHE design is frequently disregarded in the existing research. This study thus addresses an essential knowledge gap by demonstrating how user participation is critical for discovering inefficiencies and creating workable remedies. This study provides a foundation for future research that might investigate the effects of user-involved MHE design on operational efficiency, worker satisfaction, and its transferability to other sectors.

To summarise, this study highlights the significance of combining theory and practice through a mixed-methods approach to optimise MHEs. The study provides practical, human-centered solutions that improve operational efficiency and worker happiness by merging quantitative and qualitative studies. It contributes unique insights to the field of material handling systems.

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Appendix 1: Interviews

1. Head of logistics

Stakeholder involvement

Understanding stakeholder involvement was crucial for gathering valuable insights and ensuring that the Fleet Optimization strategy aligned with organizational goals. By identifying key stakeholders and comprehensively understanding their roles and expectations, the research was able to address specific concerns and integrate diverse perspectives into the analysis. This approach provided a clearer view of the operational landscape and helped tailor recommendations to meet the varied needs and priorities of stakeholders, ultimately contributing to more effective and inclusive decision-making.

1. **Executive Leadership:** Who are the key executives or senior leaders directly involved in decision-making related to material handling operations?
2. **Executive Leadership:** What are the strategic priorities and expectations of the executive leadership regarding material handling optimization?
3. **Operations and Logistics:** Who oversees day-to-day operations and logistics within the warehouse, and what is their role in the material handling optimization project?
4. **Operations and Logistics:** What specific operational challenges or pain points do the operations and logistics team currently face in material handling processes?
5. **Procurement and Supply Chain:** Who is responsible for procurement and supply chain management, and how are they involved in selecting material handling equipment and technologies?
6. **Procurement and Supply Chain:** What criteria are important for the procurement team when selecting suppliers for material handling solutions?
7. **IT and Technology:** Are there IT or technology teams involved in the material handling optimization project, and if so, what is their role?
8. **IT and Technology:** How do technology teams currently support material handling processes, and what expectations do they have for technology upgrades or integrations?
9. **Warehouse Management:** Who is responsible for overall warehouse management, and how are they involved in the material handling optimization project?
10. **Warehouse Management:** What are the warehouse management team's expectations for improvements in material handling processes?
11. **Safety and Compliance:** Are there specific stakeholders responsible for safety and compliance in material handling processes?
12. **Safety and Compliance:** What safety and compliance standards are crucial for material handling operations, and how are they currently monitored?
13. **Finance:** Who in the finance department is involved in budgeting and financial decisions related to material handling optimizations?
14. **Finance:** What financial considerations and expectations are important for the finance team in the material handling optimization project?
15. **Human Resources:** Are there HR representatives involved in the project, and what role do they play in workforce management related to material handling?

16. **Human Resources:** How is workforce training and skill development addressed in the context of material handling changes, and what are HR's expectations in this regard?
17. **Cross-Functional Collaboration:** How does cross-functional collaboration take place among these different stakeholders, and are there regular forums for coordination and communication?
18. **Cross-Functional Collaboration:** What mechanisms are in place to address conflicts or differing priorities among various stakeholders involved in the project?
19. **Decision-Making Process:** How are decisions related to material handling optimization typically made within the organization, and what is the decision-making process?
20. **Decision-Making Process:** Are there specific decision points or milestones within the project where key approvals or decisions need to be made by stakeholders?
21. **Expectations and Success Criteria:** What are the expectations and success criteria of each stakeholder group regarding the material handling optimization project?
22. **Expectations and Success Criteria:** How will the success of the material handling optimization project be measured and evaluated by different stakeholders?

Cost reduction objectives

Discussing cost-saving goals associated with the Fleet Optimization helped gathering valuable information and insights. These questions aimed to uncover specific details about the current state of operations and identify areas where cost-saving opportunities may exist. The responses guided the analysis and helped formulate recommendations tailored to goals and expectations.

23. **Operational Costs:** What are the major components of current operational costs related to material handling?
24. **Operational Costs:** Can you provide a breakdown of expenses, including labor, maintenance, fuel, and other relevant costs?
25. **Downtime Reduction:** How does downtime impact operations, and what is the current frequency and duration of downtime events?
26. **Downtime Reduction:** Are there specific factors contributing to downtime, such as maintenance issues, equipment failures, or inefficient workflows?
27. **Resource Optimization:** How are resources currently allocated in the warehouse for material handling tasks?
28. **Resource Optimization:** Are there peak periods where resources are underutilized or overutilized?
29. **Resource Optimization:** What strategies are currently in place to optimize the use of human and equipment resources?
30. **Technology Integration:** Are there existing technologies in place for material handling, and how are they contributing to operational efficiency?
31. **Technology Integration:** Are there specific technologies or automation solutions you are considering to enhance resource optimization?
32. **Equipment Utilization:** How efficiently are the current material handling vehicles utilized in day-to-day operations?
33. **Equipment Utilization:** Can you provide insights into the usage patterns of different types of equipment?
34. **Maintenance Practices:** What is the current approach to equipment maintenance, and how does it impact overall operational costs?
35. **Maintenance Practices:** Are there preventive maintenance measures in place to minimize unexpected breakdowns?
36. **Energy Efficiency:** How energy-efficient are the current material handling processes, and are there opportunities for improvement?

37. **Energy Efficiency:** Have you considered alternative energy sources or technologies to reduce energy consumption?
38. **Training and Skill Development:** How are the staff trained for material handling tasks, and are there ongoing training programs?
39. **Training and Skill Development:** Is there a correlation between skill levels and operational efficiency?
40. **Supply Chain Visibility:** How well do you have visibility into the entire supply chain, and how does it impact decision-making for material handling?
41. **Supply Chain Visibility:** Are there areas where improved visibility could lead to better resource optimization?
42. **Performance Metrics:** What key performance indicators (KPIs) are currently used to measure the efficiency of material handling operations?
43. **Performance Metrics:** Are there specific targets or benchmarks for these metrics that you aim to achieve through optimization?

Expansion plans

Examining the warehouse expansion plans was essential for gathering detailed information on the timeline, anticipated increase in storage capacity, and alignment with the broader business strategy. This inquiry aimed to provide a thorough understanding of the expansion's scope and objectives, ensuring that the material handling optimization project is effectively aligned with the organization's strategic goals. By integrating insights from these plans, the project can be better tailored to support future growth and operational demands.

1. **General Expansion Plans:** Can you provide an overview of the current status of the warehouse and the factors driving the decision to expand?
2. **General Expansion Plans:** What is the strategic objective or business goal that the warehouse expansion aims to achieve?
3. **Timeline for Expansion:** What is the proposed timeline for the warehouse expansion project?
4. **Expected Increase in Storage Capacity:** What is the targeted increase in storage capacity with the warehouse expansion?
5. **Expected Increase in Storage Capacity:** Will the expansion involve the adoption of new storage technologies or configurations to optimize space utilization?
6. **Alignment with Business Strategy:** How does the warehouse expansion align with the overall business strategy of the organization?
7. **Alignment with Business Strategy:** Are there specific market trends, customer demands, or industry changes driving the decision to expand the warehouse?
8. **Alignment with Business Strategy:** Will the expansion facilitate the introduction of new products or services, or is it primarily aimed at accommodating increased demand for existing products?
9. **Flexibility and Adaptability:** Has the expansion plan been designed with flexibility to accommodate future changes or developments?
10. **Flexibility and Adaptability:** Are there contingency plans in place for potential challenges or disruptions during the expansion process?
11. **Stakeholders Involvement:** Which stakeholders are actively involved in the decision-making and execution of the warehouse expansion?
12. **Stakeholders Involvement:** How are employees and teams within the organization being involved in the expansion process?
13. **Regulatory Considerations:** Are there any regulatory approvals or compliance requirements associated with the warehouse expansion, and how are these being addressed?

Operational challenges

Identifying current challenges or pain points in material handling processes was crucial for tailoring solutions to address specific issues effectively. Gathering insights directly from those involved in day-to-day operations provided a clear understanding of the operational difficulties faced, allowing for targeted recommendations that address these challenges. This approach ensured that the solutions developed were relevant and practical for improving overall material handling efficiency.

1. **General Operational Challenges:** Can you describe the current material handling processes within the warehouse?
2. **General Operational Challenges:** What are the primary goals or objectives of the material handling processes in the warehouse?
3. **Specific Material Handling Challenges:** What are the main challenges you face in terms of order fulfillment speed and accuracy?
4. **Specific Material Handling Challenges:** Are there specific bottlenecks or inefficiencies in the current material handling workflow?
5. **Specific Material Handling Challenges:** How do you currently manage and handle peak demand periods or fluctuations in order volumes?
6. **Equipment and Technology Challenges:** What challenges do you encounter with the current material handling equipment, if any?
7. **Equipment and Technology Challenges:** Are there technology-related challenges in the current material handling systems, such as warehouse management systems or tracking technologies?
8. **Staffing and Training Challenges:** Do you face any challenges related to staffing levels or workforce availability for material handling tasks?
9. **Staffing and Training Challenges:** How are employees trained for material handling tasks, and are there any challenges in terms of training programs or skill development?
10. **Communication and Coordination Challenges:** Are there communication challenges between different departments or teams involved in material handling?
11. **Communication and Coordination Challenges:** What other departments do you collaborate with? how do you coordinate with them and are there challenges in this coordination?
12. **Safety and Compliance Challenges:** What safety challenges or concerns exist in the current material handling processes, and how are they addressed?
13. **Safety and Compliance Challenges:** Are there compliance challenges with regulatory requirements in material handling, and how are these addressed?
14. **Customer Service Challenges:** Have there been any challenges or concerns raised by customers related to material handling, such as order accuracy or delivery times?
15. **Future Needs and Growth Challenges:** Considering the planned warehouse expansion, do you foresee any challenges in scaling up material handling processes?
16. **Feedback Mechanisms:** How do you currently gather feedback from employees regarding material handling processes, and have any recurring issues been identified?
17. **Feedback Mechanisms:** Are there compliance challenges with regulatory requirements in material handling, and What channels or processes are in place for reporting and addressing issues or challenges related to material handling?
18. **Your Own Experience:** Are there additional insights or specific challenges that haven't been covered, but you believe are important to address in the context of material handling?
19. **Your Own Experience:** Are there compliance challenges with regulatory requirements in material handling, and What channels or processes are in place for reporting and addressing issues or challenges related to material handling?

Efficiency improvement goals

Exploring the efficiency improvement goals and specific metrics for material handling operations aimed to uncover both overarching objectives and crucial performance indicators. This inquiry focused on understanding the desired improvements in efficiency and the metrics used to measure success. The gathered responses served as a foundation for developing targeted optimization strategies that align with these goals and address the specific needs of the warehouse operations.

1. **Efficiency Improvement Objectives:** What are the primary objectives you aim to achieve through the improvement of material handling efficiency in the warehouse?
2. **Efficiency Improvement Objectives:** How do you envision the impact of enhanced material handling efficiency on the overall warehouse operations?
3. **Efficiency Improvement Objectives:** Are there specific pain points or challenges in the current material handling processes that are driving the need for efficiency improvement?
4. **Efficiency Improvement Objectives:** What role does material handling efficiency play in achieving broader organizational goals and strategies?
5. **Specific Efficiency Metrics:** Can you specify the key efficiency metrics you are looking to enhance within the material handling operations?
6. **Specific Efficiency Metrics:** Are there particular aspects of order fulfillment speed that you aim to improve, and what benchmarks are you targeting?
7. **Specific Efficiency Metrics:** When it comes to inventory turnover, what level of improvement or optimization are you aiming for?
8. **Specific Efficiency Metrics:** How would you define and measure resource utilization within the context of material handling, and what improvements are sought in this area?
9. **Specific Efficiency Metrics:** Are there any other efficiency-related metrics or KPIs that are particularly important for your organization in the context of material handling?
10. **Specific Efficiency Metrics:** Can you provide historical data or benchmarks for the current performance of these efficiency metrics in material handling?
11. **Specific Efficiency Metrics:** What specific benefits do you anticipate from achieving improvements in order fulfillment speed, inventory turnover, and resource utilization?

Technology adoption objectives

Exploring the willingness and interest in adopting new technologies for material handling aimed to uncover the current stance, preferences, and considerations regarding technological advancements. This inquiry focused on understanding the openness to new technologies, the types of technologies of interest, and any associated concerns or criteria for adoption. The gathered responses provided a foundation for evaluating potential technological solutions and aligning them with the organization's operational needs and strategic goals.

1. **General Technology Adoption:** How would you describe the current level of technology adoption in your material handling processes?
2. **General Technology Adoption:** What role do you see technology playing in the future of material handling within your organization?
3. **Automation and Robotics:** Are you open to adopting automation and robotics in material handling processes, and if so, in which specific areas?
4. **Automation and Robotics:** What expectations or concerns do you have regarding the implementation of automated material handling solutions?
5. **Automation and Robotics:** Have you conducted any feasibility studies or pilot programs related to automated material handling?
6. **Internet of Things (IoT):** Are you interested in integrating IoT technologies for real-time tracking and monitoring of material handling equipment and inventory?

7. **Internet of Things (IoT):** What specific challenges or considerations do you see in adopting IoT solutions for material handling?
8. **Internet of Things (IoT):** How do you envision IoT contributing to the optimization of material handling processes and overall warehouse efficiency?
9. **Data Analytics:** Is there an interest in leveraging data analytics for predictive maintenance, performance optimization, or demand forecasting in material handling?
10. **Data Analytics:** What data sources do you currently have in place for material handling, and how are they utilized?
11. **Data Analytics:** What data analytics tools or platforms, if any, are you currently using or considering for material handling optimization?
12. **Technology Integration Challenges:** Are there specific challenges or barriers you anticipate in integrating new technologies into your material handling processes?
13. **Technology Integration Challenges:** What level of customization or flexibility do you expect from technology providers to meet your specific material handling requirements?
14. **Return on Investment (ROI):** How do you evaluate the return on investment (ROI) when considering the adoption of new technologies for material handling?*)*
15. **Return on Investment (ROI):** What timeframes do you typically consider for realizing returns on technology investments in material handling?
16. **Supplier Collaboration:** Do you prefer working with technology suppliers who offer end-to-end solutions, or do you prefer integrating best-of-breed technologies from multiple suppliers?
17. **Supplier Collaboration:** What criteria do you consider when selecting technology suppliers for material handling solutions?
18. **Training and Change Management:** How do you plan to address training needs and change management associated with the adoption of new technologies in material handling?
19. **Regulation and Compliance:** Are there regulatory or compliance considerations that may impact the selection and adoption of certain technologies in material handling?
20. **Future Technological Trends:** Are there specific technological trends or innovations you are monitoring for potential adoption in material handling in the future?

Regulatory and compliance considerations

Exploring regulatory and compliance considerations for the design and implementation of material handling solutions aimed to ensure that proposed changes align with legal standards and industry regulations. This inquiry focused on understanding the specific regulatory requirements and compliance standards that must be met during the development and deployment of material handling solutions. The gathered responses provided critical insights into the legal and regulatory framework guiding the project, ensuring that all proposed solutions adhere to necessary standards and mitigate any risks associated with non-compliance.

1. **Industry-Specific Regulations:** What industry-specific regulations or standards govern material handling operations in your sector?
2. **Industry-Specific Regulations:** How is compliance with these industry-specific regulations currently ensured in your material handling operations?
3. **Local and National Regulations:** Are there specific local or national regulations that apply to material handling practices in your geographic location?
4. **Local and National Regulations:** How is compliance with local and national regulations monitored and reported in your current material handling processes?
5. **Environmental Regulations:** Are there environmental regulations that impact material handling processes, such as waste disposal or emissions standards?

6. **Environmental Regulations:** How is the environmental impact of material handling operations currently assessed and addressed?
7. **Safety Standards:** What safety standards and regulations apply to material handling equipment and processes in your industry?
8. **Safety Standards:** How are safety protocols and compliance with safety standards currently integrated into material handling operations?
9. **Worker Safety and Ergonomics:** Are there specific regulations related to worker safety and ergonomics in material handling roles?
10. **Worker Safety and Ergonomics:** How is worker safety and ergonomic compliance currently addressed in your material handling processes?
11. **Reporting and Documentation:** What reporting and documentation requirements exist for material handling processes as per regulatory standards?
12. **Reporting and Documentation:** How are records of compliance maintained, and what documentation is currently required to meet regulatory standards?
13. **Changes in Regulations:** Are there upcoming changes in regulations that may impact material handling practices, and how do you stay informed about these changes?
14. **Changes in Regulations:** What mechanisms are in place to adapt material handling processes to changes in regulatory requirements?
15. **Third-Party Certifications:** Do you pursue or hold any third-party certifications related to material handling operations and compliance?
16. **Third-Party Certifications:** How do third-party certifications impact your material handling processes, and do they influence decision-making in equipment selection?
17. **Employee Training and Awareness:** How are employees trained and made aware of regulatory requirements in material handling roles?
18. **Employee Training and Awareness:** Are there ongoing initiatives to keep employees informed about changes in regulations affecting their work?
19. **Contingency Planning:** In the event of non-compliance or changes in regulations, what contingency plans are in place to address potential challenges?
20. **Contingency Planning:** How does the organization ensure that material handling processes remain compliant during periods of regulatory transition or uncertainty?

Performance metrics

Discussing the key performance indicators (KPIs) to measure the success of material handling operations aimed to identify the metrics crucial for assessing operational efficiency and effectiveness. This inquiry focused on understanding which performance indicators are most significant for evaluating the success of material handling processes. The gathered responses provided insights into the essential KPIs that guide performance measurement, enabling the development of strategies that are aligned with operational goals and performance objectives.

1. **Current Key Performance Indicators (KPIs):** What are the current key performance indicators (KPIs) that you use to assess the overall performance of material handling operations in the warehouse?
2. **Current Key Performance Indicators (KPIs):** How do these KPIs align with the broader organizational goals and objectives related to material handling efficiency?
3. **Operational Efficiency Metrics:** What specific metrics do you use to evaluate the efficiency of order fulfillment processes within the warehouse?
4. **Operational Efficiency Metrics:** How is inventory turnover measured, and are there specific KPIs related to optimizing inventory management in material handling?
5. **Resource Utilization Metrics:** Do you track resource utilization metrics, such as the efficiency of material handling equipment and workforce productivity?

6. **Resource Utilization Metrics:** Are there specific KPIs related to minimizing downtime of material handling equipment and maximizing their availability?
7. **Cost-Related Metrics:** What cost-related KPIs are crucial for assessing the economic efficiency of material handling operations?
8. **Cost-Related Metrics:** How do you evaluate the cost-effectiveness of material handling processes, and are there specific financial KPIs you focus on?
9. **Customer Service Metrics:** Are there customer-centric KPIs related to material handling, such as on-time delivery, order accuracy, or customer satisfaction scores?
10. **Customer Service Metrics:** How do you measure the success of material handling operations in meeting customer expectations and demands?
11. **Safety and Compliance Metrics:** What metrics are used to assess the safety performance of material handling processes, and how is compliance with safety standards measured?
12. **Safety and Compliance Metrics:** Are there specific regulatory or compliance KPIs that impact material handling operations, and how are these monitored?
13. **Continuous Improvement Metrics:** How do you measure and track continuous improvement in material handling processes, and are there KPIs specifically dedicated to improvement initiatives?
14. **Continuous Improvement Metrics:** What mechanisms are in place to monitor and evaluate the effectiveness of implemented process improvements in material handling?
15. **Environmental and Sustainability Metrics:** Are there metrics related to environmental sustainability in material handling, such as energy efficiency or waste reduction?
16. **Environmental and Sustainability Metrics:** How do you measure the environmental impact of material handling operations, and are there specific KPIs for sustainability goals?
17. **Technology Adoption Metrics:** In the context of technology adoption, are there KPIs that measure the success and impact of technology integration in material handling processes?
18. **Technology Adoption Metrics:** How are data analytics and technology-driven insights reflected in KPIs related to material handling performance?
19. **Employee Satisfaction and Training Metrics:** Do you measure employee satisfaction or engagement specific to material handling roles, and are there KPIs related to workforce morale?
20. **Employee Satisfaction and Training Metrics:** How are training and skill development measured in the context of material handling, and are there KPIs for assessing the effectiveness of training programs?
21. **Next Steps and Adaptability:** Considering the warehouse expansion, are there new KPIs or adaptations to existing KPIs that are envisioned for the future state of material handling operations?
22. **Next Steps and Adaptability:** How often are these KPIs reviewed and updated, and what mechanisms are in place for adapting them to changing business needs?

Supplier relationship expectations

Discussing the expectations from suppliers aimed to clarify the critical factors that influence the effectiveness of supplier partnerships, including lead times, customization options, and ongoing support. This inquiry focused on understanding the key criteria for evaluating supplier performance and the requirements that must be met to ensure successful collaboration. The gathered responses provided valuable insights into the standards and expectations for supplier relationships, helping to align supplier capabilities with organizational needs and enhance the overall effectiveness of the supply chain.

1. **Lead Times:** What is the typical lead time you expect from suppliers for delivering material handling equipment or technologies?
2. **Lead Times:** Are there any specific considerations that may influence your acceptable lead times for material handling equipment?
3. **Customization Options:** To what extent do you require customization options for material handling equipment to meet your specific operational needs?
4. **Customization Options:** Could you provide examples of features or specifications that you may want to customize in material handling equipment?
5. **Quality and Reliability:** What quality standards or certifications are essential for material handling equipment suppliers to meet your expectations?
6. **Quality and Reliability:** How critical is the reliability and durability of material handling equipment in your decision-making process?
7. **Cost Considerations:** What is your approach to balancing cost considerations when selecting material handling equipment suppliers?
8. **Cost Considerations:** Do you prefer long-term contracts or agreements with suppliers to secure stable pricing, or are you open to other pricing models?
9. **Ongoing Support and Maintenance:** What level of ongoing support and maintenance services do you expect from material handling equipment suppliers after the purchase?
10. **Ongoing Support and Maintenance:** How crucial is the availability of spare parts and the responsiveness of suppliers in addressing maintenance issues?
11. **Communication and Collaboration:** What communication channels and frequency do you prefer with material handling equipment suppliers during the procurement and implementation phases?
12. **Communication and Collaboration:** How do you envision collaboration with suppliers in addressing challenges or improvements in material handling processes over time?
13. **Sustainability and Environmental Considerations:** To what extent do you consider sustainability and environmental factors when selecting material handling equipment suppliers?
14. **Sustainability and Environmental Considerations:** Are there specific environmental certifications or initiatives you expect from suppliers in the material handling industry?
15. **Flexibility and Adaptability:** How important is supplier flexibility and adaptability in responding to changing needs or unexpected challenges in the material handling environment?
16. **Flexibility and Adaptability:** What mechanisms do you prefer for addressing changes in material handling requirements or unforeseen circumstances with suppliers?
17. **Feedback Mechanisms:** How do you provide feedback to suppliers, and what expectations do you have for them to address and act upon your feedback?
18. **Contractual and Legal Aspects:** Are there specific contractual or legal considerations that are particularly important to you in agreements with material handling equipment suppliers?
19. **Future Collaboration:** In the context of future collaboration, what are your expectations for scalability, additional features, or upgrades from material handling equipment suppliers?
20. **Future Collaboration:** What role do you see technology advancements playing in your ongoing relationship with material handling equipment suppliers?

Timeline and urgency

Gathering information about timeframes and priorities for implementing changes aimed to clarify the schedule and urgency associated with material handling improvements. This inquiry focused on understanding the critical deadlines and priority levels for various changes, as well as any urgent requirements that need to be addressed promptly. The gathered responses provided a detailed view of the expected implementation timelines and the urgency of different tasks, enabling the development of a well-structured plan that aligns with organizational priorities and ensures timely execution of necessary improvements.

1. **Overall Project Timeline:** What is the overall timeline for implementing changes in the material handling processes, considering the warehouse expansion and fleet optimization project?
2. **Overall Project Timeline:** Are there specific milestones or deadlines within the project timeline that are particularly critical or time-sensitive?
3. **Lease Contract Expiration:** With the lease contract expiring in 2025, what implications or deadlines does this create for the implementation of changes in material handling?
4. **Lease Contract Expiration:** Are there any contractual obligations or considerations related to the lease contract expiration that impact the timing of material handling optimizations?
5. **Urgent Requirements:** Are there any urgent or critical material handling challenges that need to be addressed immediately, irrespective of the overall project timeline?
6. **Urgent Requirements:** In the context of urgency, are there specific areas within material handling that require prioritized improvements or modifications?
7. **Implementation Phases:** How is the implementation of changes in material handling planned to be phased over time, and are there phases with higher urgency?
8. **Implementation Phases:** Are there dependencies between different aspects of the material handling optimization project that could impact the overall timeline?
9. **Stakeholder Expectations:** How do stakeholders within the organization perceive the urgency of material handling optimizations, and are there divergent views on the timeline?
10. **Stakeholder Expectations:** What mechanisms are in place to communicate and manage expectations regarding the project timeline with relevant stakeholders?
11. **Regulatory and Compliance Deadlines:** Are there any regulatory or compliance deadlines that need to be considered in the implementation of changes to material handling processes?
12. **Regulatory and Compliance Deadlines:** How are regulatory or compliance requirements integrated into the project timeline, and are there deadlines that must be met?
13. **Resource Availability:** How does resource availability, both internal and external, impact the project timeline for material handling optimizations?
14. **Resource Availability:** Are there any external factors, such as supplier lead times, that may influence the timeline for acquiring new material handling equipment?
15. **Contingency Planning:** Is there a contingency plan in place to address unexpected delays or challenges that may arise during the implementation of material handling changes?
16. **Contingency Planning:** How are potential risks and uncertainties factored into the overall timeline, and what steps are taken to mitigate them?
17. **Decision-Making Process:** What is the decision-making process for approving and implementing changes in material handling, and how might this impact the timeline?
18. **Decision-Making Process:** Are there specific decision points or milestones within the project where key approvals or decisions need to be made that impact the timeline?

2. Logistics Manager

1. How do material handling practices impact transportation and logistics efficiency within the warehouse?
2. Are there specific bottlenecks or challenges in material handling that affect overall logistics coordination?
3. In your experience, how can material handling processes be optimized to streamline logistics operations?
4. What information or data from material handling teams is crucial for effective logistics planning?
5. Are there specific requirements or expectations you have for material handling to enhance overall logistics efficiency?
6. In your view, how can technology be integrated into material handling processes to improve coordination with logistics activities?
7. How are employees trained for material handling tasks and equipment operation?

8. How do employees typically communicate regarding material handling challenges or issues?
9. From your perspective, are there any specific areas that could be improved within material handling operations?
10. Is there a formalized system for employees to provide feedback on material handling processes or equipment?

3. Warehouse operations manager

1. How would you describe the current state of material handling operations in terms of efficiency and effectiveness?
2. What are the key performance indicators you use to assess the success of material handling processes?
3. From a managerial perspective, what challenges do you see in optimizing material handling workflows?
4. Are there specific goals or expectations you have in mind for the improvement of material handling operations?
5. How do you envision the role of technology in enhancing material handling efficiency within the warehouse?
6. In your opinion, what are the critical areas that should be addressed to achieve optimal material handling performance?
7. How are employees trained for material handling tasks and equipment operation?
8. How do employees typically communicate regarding material handling challenges or issues?
9. From your perspective, are there any specific areas that could be improved within material handling operations?

4. Warehouse safety manager

1. What are the common issues or challenges you encounter in the maintenance of material handling equipment?
2. In your experience, how does the current material handling setup contribute to wear and tear on equipment?
3. Are there specific suggestions for equipment upgrades or replacements that could improve overall reliability and longevity?
4. How do routine maintenance schedules align with material handling workflows, and are there areas for improvement?
5. What role does technology play in facilitating equipment maintenance within the context of material handling?
6. In your opinion, how can the material handling setup be optimized to reduce the frequency of maintenance issues and improve overall equipment performance?
7. How do material handling practices align with safety regulations in the warehouse?
8. Are there specific safety concerns related to material handling that need immediate attention or improvement?
9. In your experience, how can safety protocols be better integrated into material handling workflows?
10. What role does employee training and awareness play in ensuring safety during material handling activities?
11. Are there suggestions for enhancing safety measures without compromising material handling efficiency?

12. How can technology, such as sensors or monitoring systems, contribute to improving safety in material handling?
13. How are employees trained for material handling tasks and equipment operation?
14. How do employees typically communicate regarding material handling challenges or issues?
15. From your perspective, are there any specific areas that could be improved within material handling operations?
16. Is there a formalized system for employees to provide feedback on material handling processes or equipment?

Appendix 2: Introduction to HAMC's logistics operations

The initial objective was to provide a thorough introduction to HAMC's warehouse processes and operations, offering a comprehensive understanding of the broader logistics framework and organizational structure, as presented in figure 7. Engagements started with crucial departments such as quality operations, transport operations, and warehouse operations.

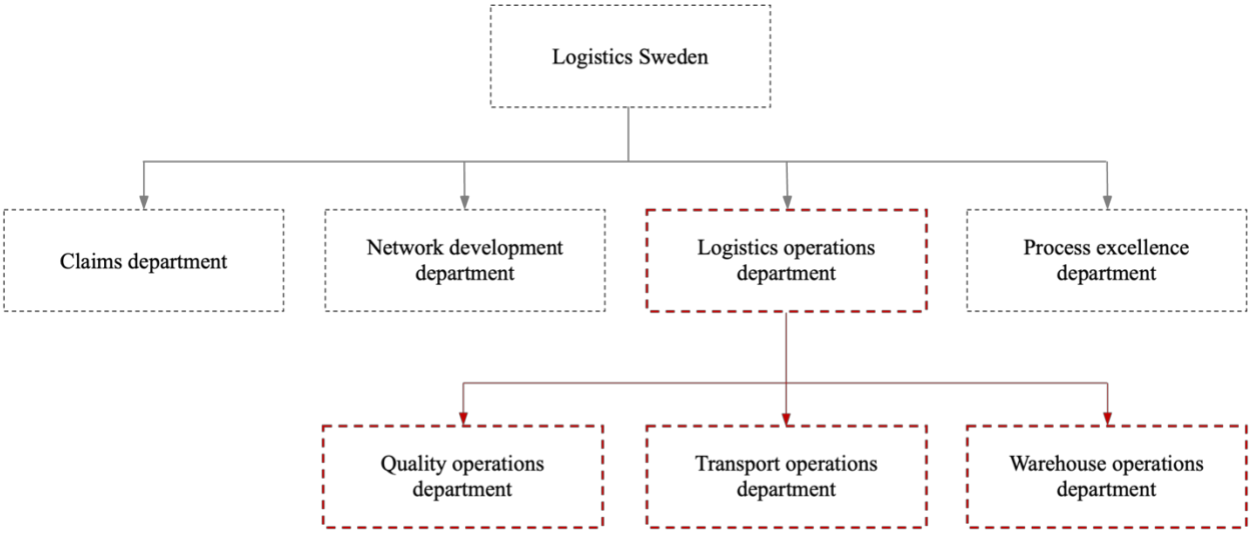


Figure 1. HAMC's logistics organizational structure in Sweden. Source: HAMC

Quality operations department

The quality operations department monitors and ensures that all operations at HAMC's warehouse meet quality standards and requirements. They do regular checks and fix any problem right away in order to keep everything run smoothly. Furthermore, providing training to colleagues on proper procedures to guarantee compliance with safety and quality protocols. They are continually seeking for methods to improve operations and are committed to continuous improvement. A detailed presentation was held about quality standards, regulatory compliance, and avoidance of operational risks, including disruptions, of MHE so that the required benchmarks could be understood. They provided valuable insights on issues like reliability, durability, and maintenance needs. The possibility of problems occurring after implementation of the new MHE was reduced thanks to their participation in the selection process, which enabled the early detection and mitigation of any risk. In the end, their input ensured that the warehouse's equipment selections were in line with quality standards, regulatory requirements, and operational needs, resulting in smoother warehouse operations.

Transport operations department

All goods entering and leaving HAMC's warehouse are closely monitored by the transport operations department. The administration of loads, transportation schedules, costs, customer service, and performance evaluations are all part of their responsibilities. In order to fulfil delivery dates, they make sure that shipments are booked successfully, that loads are safely and efficiently handled to minimize damage and maximize space utilization, and that items are delivered via the most ideal routes. In order to pinpoint problem areas, a detailed presentation was given on critical transportation indicators and performance analysis. Their involvement in the process ensures that equipment selection meets all of the transportation demands and standards, works with what is

already in place, reduces the risk of accidents and injuries, shortens travel times, and provides the best value in terms of performance and cost-effectiveness.

Warehouse operations department

The warehouse operations department manages the day-to-day activities within HAMC's warehouse. Among their many responsibilities is the improvement of workflow efficiency, facility layout design, storage optimization, order fulfilment, and inventory management. They are responsible for the effective and timely receipt, storage, and dispatch of goods in order to satisfy client needs. They are also responsible for making sure the warehouse's facilities and equipment are well-maintained and in good working condition to meet operational needs. A detailed presentation covering warehouse layout, inventory management processes, order processing workflows, and facility maintenance protocols was conducted to gain insights into the operational requirements for MHE. Their involvement in the process ensured that equipment selection works well with the rest of the warehouse's processes, optimizes storage space utilization, simplifies order fulfillment, and enhances overall workflow efficiency.

Appendix 3: Overview of HAMCs informatic systems

HAMC relies on several informatic systems to manage its operations within the warehouse. Some of these that are relevant to the project are SAP, Transporeon, and 3PMHES Connect.

Sap

The SAP system is an enterprise resource planning (ERP) software which integrates different business activities and processes into a single platform. HAMC uses SAP as its main platform to handle many aspects of its warehouse operations. SAP has modules for inventory management, sales, finance, human resources, and more.

HAMC mainly uses SAP for inventory management, to keep track, monitor and control its inventory in real time, monitor and regulate its movement, and optimize it to meet customer needs. On top of that, SAP makes order processing easier, which means HAMC can handle client orders, make picking lists, and organize order fulfilment.

On a daily basis, HAMC simplified operational operations with the help of SAP's Warehouse Management System (WMS), which offers functions including bin management, picking and packing, put-away, and shipping. In addition, HAMC gets insight into its performance measures by using SAP's analytics and reporting solutions. With the use of advanced analytics, HAMC assesses order fulfilment efficiency, inventory turnover rates, and other critical metrics to find ways to improve.

Transporeon

The Transporeon system is a cloud-based platform used by logistic companies that helps them with transport processes. HAMC uses Transporeon to handle transportation operations such as planning, execution, and monitoring of transportation related activities.

Transporeon features modules for carrier management, freight tendering, time slot management, and real-time tracking. With these, HAMC is able to oversee its transportations. Through Transporeon, HAMC assigns carriers, tenders freight, and schedules deliveries, ensuring that orders are transported to customers on time and at the lowest cost.

Transporeon offers analytics and reporting capabilities, providing insights into on-time delivery rates, carrier performance, and transportation costs. HAMC uses these features to improve its transportation operations for efficiency and customer satisfaction.

3PMHES Connect

3PMHES developed the 3PMHES Connect system, a digital platform for MHE management. HAMC relies on 3PMHES Connect to manage its MHE fleet; it ensures everything runs smoothly, safely and productively. With the help of 3PMHES Connect, HAMC is able to optimize its fleet, preform predictive maintenance, and remotely monitor its MHE fleet, allowing for proactive fleet management. HAMC could use real-time data on consumption and performance from 3PMHES Connect to spot problems before they escalate and plan preventative maintenance.

3PMHES Connect makes it easier to distribute and employ MHE resources in the warehouse efficiently. This ensures that the correct equipment is deployed to the right tasks at the right time. Workflow efficiency and operating expenses are both enhanced.

In addition, 3PMHES Connect provides tools to track operators' performance and safety compliance, which helps to promote a culture of safety and accountability within the warehouse. By tracking operator behavior and adherence to safety protocols, HAMC can minimize the risk of accidents and injuries, creating a safer working environment for warehouse personnel.

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